The two finite chiral atlases: recovered results and source corrections

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This manuscript gives a source-wide disposition of Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Mathematica Contemporanea 5 (2012), 371–382. The first atlas and all printed entries of the second atlas have complete accepted finite evidence. Several printed numbers and one automorphism-group label require correction. The larger domain stated in Section 4 also requires an explicit boundary: the printed roster has 79 groups and omits one group satisfying that section's literal condition. We do not invent a census for that omitted group.

The appended complete component proofs and the exact source ledger specify what is acquired. Standard regular and chiral reconstruction theorems retain the source's attribution to McMullen–Schulte and Schulte–Weiss. Exact finite GAP group operations, the named group constructions and automorphism classifications, and completeness of the earlier regular-cover atlas remain stated inputs. No present website is asserted to have identical historical bytes. This synthesis repeats no group census.

Definitions and equivalence. We use the source's ranked-poset definition, including the diamond axiom and strong flag connectivity. Its Sections 2–3 group criteria are retained with their complete subset intersection property. The accepted prefix-coordinate manuscript supplies the interval/difference subgroup dictionary, reflection criterion and complete stabilizer enumeration; it does not replace polytope recognition by adjacent product orders alone. A colour-preserving isomorphism of flag graphs is a polytope isomorphism. Reversing all colours gives duality, which is not additionally identified in the atlas counts below.

In the first atlas, the two enantiomorphic markings are identified as one unrooted chiral polytope. In the second atlas, a generating rotation marking is taken modulo the full automorphism group of its abstract group; its mirror is counted separately. The reflection criterion implies that this mirror involution has no fixed chiral class. Thus division by two there forgets orientation, not duality. Regular columns count full regular automorphism groups, not rotation groups extracted from the chiral search.

Theorem 1 (Theorem 3.1 and the source algorithm)**.** *Let the actual flag graph of a quotient polytope be the right-coset graph of $N \leq W = \langle r_0, \ldots, r_{d-1} \rangle$, where the r_i are its specified flag adjacencies. Its colour automorphism group is $N_W(N)/N$. The quotient is chiral precisely when $[W : N_W(N)] = 2$ and all r_i lie outside $N_W(N)$. There is at most one index-two subgroup H avoiding all r_i. For that H, the possible chiral stabilizers are precisely $N \lhd H$ not normal in W, subject to the actual polytope condition. Conjugate stabilizers N and $r_0 N r_0$ describe the same unrooted quotient.*

Proof. A colour automorphism is determined by the image Na of N and must send Nw to Naw. This is well-defined exactly when a normalizes N. The resulting action is free, with $[W : N_W(N)]$ orbits. For an index-two normalizer, adjacency by r_i exchanges the two orbits exactly when r_i is outside it. The subgroup avoiding every generator is the kernel of the unique possible parity homomorphism $r_i \mapsto 1 \in C_2$. A subgroup normal in that H but not in W has normalizer exactly H, so it has the two stated conjugates. The complete coset and geometric interface proofs are in *Chiral quotients from one parity subgroup.* □

The source's core-free filter is optional: the kernel of the flag action is $\mathrm{Core}_W(N) = N \cap r_0 N r_0$. It is not legitimate to discard nontrivial cores when excluding arbitrary chiral quotients. Nor do we infer that every monodromy quotient is a regular polytope. Minimal faithful-cover language is used only with its specified flag-action interface.

Theorem 2 (The first atlas and Table 1)**.** *Relative to the acquired complete regular-cover atlas of orders at most 2000, excluding 1024 and 1536, there are exactly 48 rank-three and eight rank-four chiral quotients, with distinct duals counted separately. The four Table 1 markings and their reversals are the eight rank-four examples. Their displayed cover and rotation-group structures are correct. The vertex figures of the last two rows have types* $\{6, 18\}$ *and* $\{6, 9\}$ *and are among the verified rank-three chiral examples.*

Proof. The accepted full source parser supplies 9,212 nondegenerate marked covers. The acquired product argument excludes a bond of order two; ranks one and two cannot be chiral. All normal stabilizers allowed by the preceding criterion are considered, including nontrivial cores. Three exact completed intervals partition the source roster. Their 8,986 normal candidates leave 756 after the necessary base-face intersection test.

Literal coset transport verifies 704,608 edges. Of the 756 candidates, 360 have full colour isomorphisms to the 56 accepted examples, covering all of them; the other 396 have explicit negative witnesses, namely 350 diamond failures and 46 disconnected flag residues. The examples themselves have complete face-poset, chain, diamond and residue checks, and complete rooted colour comparisons proving distinctness and chirality. This is the accepted *Identifying the quotient flag graphs* component, with no reliance on an interrupted candidate or on abstract database identifiers for exclusions.

Table 1's six structural names were proved on the actual cover groups and rotation quotients by literal generator-edge transports, normal subgroups, complements and commuting factors. In particular the last two rows really have the same named cover and rotation groups. The source convention is $|D_9| = 18$; its parentheses are preserved. The complete names and certificates are in *The six group descriptions in the small chiral atlas*.

For the remaining vertex-figure sentence, keep the base vertex in either accepted rank-four flag graph and take its component using colours $1, 2, 3$. Renumbering those colours gives the vertex-figure flag graph by the polytope section definition. The saved components have respectively 216 and 108 flags. Complete bijections to accepted rank-three entries 44 and 39 preserve all 648 and 324 coloured edges. Thus the sections are the asserted chiral polytopes. Every vertex has flags from both global flag orbits (an adjacency of colour one preserves the vertex and exchanges those orbits), so the chiral automorphism group is vertex-transitive. The base-vertex calculation therefore applies to all vertex figures of each example. □

Theorem 3 (All printed almost-simple entries)**.** *For the 79 distinct named group models in Tables 2–7, occurring in 88 rows, the regular and oriented chiral counts are as follows. All rows agree rank by rank except the regular rank-three entry for* $\mathrm{PSL}_2(27)$*, which is 26 rather than 27, and the chiral rank-three entry for* $\mathrm{PSU}_4(2) \cong \mathrm{PSp}_4(3)$*, which is 46 rather than 26.*

rank	3	4	5	6	7	8	*total*
regular	8319	810	61	16	3	1	9210
oriented chiral	15056	1890	28	0	0	0	16974

The oriented chiral classes form 8487 mirror pairs.

Proof. The accepted regular application acquires 14 earlier complete censuses and 65 complete exact executions. Its full automorphism actions are verified common actions on the named models; it does not use an inner action in place of the full automorphism group. The acquired records bind their actual drivers, models and outputs, with complete terminal rank lists; the

older records' runtime-file-trace limits remain explicit. A separate fixed-endpoint calculation gives $312/12 = 26$ for the exceptional regular row.

The accepted chiral application acquires 72 complete whole searches, the complete 25-branch S_9 union, and six complete fixed-first-class unions. A fixed first list is proved to contain exactly one representative of each relevant conjugacy class. Unions require its literal equality, not just matching class indices. Each centralizer recursion is complete by the acquired stabilizer theorem. Heredity and strict subgroup growth supply its all-rank stopping boundary. Full automorphisms act freely on generating markings, so division of the complete inner-orbit count by $|\mathrm{Out}(G)|$ is valid separately at each rank and type. An independent literal positive certificate already distinguishes 46 chiral classes for the exceptional unitary row, without using its census completeness or the outer-order division.

The accepted applications join these results to every printed occurrence through literal named models. Nine repeated occurrences are removed only after their group identifications; they contribute 95 regular and 50 chiral counts to the raw family sums. Their removal gives the displayed totals. No source count was a filter or premise of these enumerations. □

Family totals and the observations in Section 6. The exact family sums, retaining repeated occurrences between families, are

Table	2	3	4	5	6	7
regular	1536	804	3903	1846	1202	14
oriented chiral	4780	1710	1882	3998	4242	412

Besides the two row corrections, the printed aggregate arithmetic in Table 3 (1702), Table 4 (2050), and Table 5 (1840 regular) is corrected by these sums. Section 6's global 9205, 17114 and 8557 become respectively 9210, 16974 and 8487. Its PSL_2-family rank-at-least-four counts are 45 regular and 338 chiral, replacing 51 and 506. The comparative comments about sporadic, unitary and Suzuki groups remain statements about this finite roster and its stated convention. They do not compare all groups or all genera. As in the source's aggregate, $\mathrm{Aut}(J_2)$ is excluded: its order 1209600 lies beyond both stated cutoffs. The older 5265 regular objects identified duals; the historical Conder genus counts, MAGMA timings and website descriptions remain attributed context, not new calculations or interchangeable counting conventions.

Orders and named automorphism groups. The complete source application identifies every printed group and automorphism label through acquired constructor/classification and literal transport or outer-preimage routes. For $S \le G \le \mathrm{Aut}(S)$ with S simple, restriction identifies $\mathrm{Aut}(G) \cong N_{\mathrm{Aut}(S)}(G)$: the unique simple socle is characteristic, its centralizer is trivial, and a normalizing automorphism extends by conjugation. The actual preimages and small outer normalizers, not order equality alone, supply the names. The accepted $J_2 : 2$ join additionally has an actual outside involution complement.

The source's order 443510 for M_{22} is corrected to 443520, and its order 7800 for $\mathrm{PSL}_2(25).2$ to 15600. The automorphism label $P\Gamma L_3(3)$ for $\mathrm{PSU}_3(3)$ is corrected to $P\Gamma U_3(3)$, of order 12096. In contrast $P\Gamma L_3(3) = PGL_3(3)$ has order $(26 \cdot 24 \cdot 18)/2 = 5616$.

The historical-domain statement requires a correction. The abstract uses a 900000 cutoff and speaks of the group Γ appearing in the 1985 ATLAS. Section 4 instead gives a one-million cutoff and explicitly requires only the simple socle S to appear. Under that latter socle condition, the accepted complete simple/outer roster and the explicitly retained CTblLib transcription of the 1985 contents give 80 almost-simple isomorphism classes. Exactly 79 have order below 900000; these are precisely the printed models. The additional group is $\mathrm{Sp}_4(4)$, of order 979200, whose simple socle has ordinary ATLAS page 44. Thus the printed roster does

not exhaust the larger Section 4 domain. This corrects the coverage claim, rather than silently replacing its domain by the printed list.

The historical ATLAS transcription is an explicit premise, not a claimed direct inspection of a book scan. We do not infer that every outer extension had its own printed ATLAS table, nor what an unavailable historical program executed. No polytope count, zero result, or new census for $\mathrm{Sp}_4(4)$ is asserted. The recovered theorem is the complete printed 79-group atlas with the corrected entries, plus this exact larger-domain omission.

Attribution and disposition. The original constructions, examples and enumeration programme retain the source authors' credit. The complete component proofs identify their borrowed classification and software interfaces. The Human Compression project supplied the finite applications and operational repairs, with individual contributions, distinct reviews, the normalizer suggestion and advisory source readings recorded in the component provenance. The only additional finite join in this synthesis is the two vertex-figure maps. This is a corrected source disposition, submitted for a separate whole-source review. It claims no new conjecture solution, Lean proof, compression measurement or automatic publication status.

Chiral quotients from one parity subgroup

We recover the quotient criterion and search of Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Math. Contemp. 5 (2012), printed page 375. The underlying group-action argument is classical. Its role here is to specify one reusable interface, including the two different equivalence conventions in that paper. This note does not establish either of the paper's complete finite atlases.

Let $W = \langle r_0, \dots, r_{n-1} \rangle$ be a finite group generated by involutions, and $N \leq W$. On the right cosets Nw, put a colour-i edge from Nw to Nwr_i. Loops and coincident edges are permitted at this algebraic stage. The graph is connected because the r_i generate. When a quotient polytope is used below, we require its actual flag graph to be this graph. The source's semisparse quotient theorem supplies that geometric interface; an arbitrary orbit poset is not sufficient.

Theorem 1. *The colour-preserving automorphism group of the coset graph is $N_W(N)/N$. It acts freely, with $[W : N_W(N)]$ vertex orbits. If this is the flag graph of a polytope, that polytope is chiral if and only if $[W : N_W(N)] = 2$ and $r_i \notin N_W(N)$ for every i.*

Proof. A colour-preserving automorphism commutes with each of the permutations induced by r_i, hence with their products. If it sends N to Na, then it must send Nw to Naw. For $x \in N$, well-definedness at N requires $Nax = Na$, so $aNa^{-1} \leq N$. The two subgroups have the same finite order, hence equality. Thus $a \in N_W(N)$. Conversely any such a gives a well-defined bijection $Nw \mapsto Naw$ commuting with all colours. Two choices give the same map exactly when their cosets modulo N agree. This proves the asserted group identification.

If $Naw = Nw$, then $a \in N$, so the action is free. Its orbits therefore number

$$(|W|/|N|)/(|N_W(N)|/|N|) = [W : N_W(N)].$$

A polytope's automorphisms are precisely its colour-preserving flag-graph automorphisms: its i-faces are the components after deleting colour i, and face incidence is witnessed by flags. This uses the stipulated actual flag graph.

Write $H = N_W(N)$ and suppose $[W : H] = 2$, so $H \lhd W$. The vertices Nw and Nwr_i are in the same H/N-orbit exactly when $wr_iw^{-1} \in H$, equivalently $r_i \in H$. Consequently every adjacency exchanges the two orbits exactly when all r_i lie outside H. This is the defining flag-orbit condition for chirality. □

Proposition 2. *There is at most one subgroup H of index two avoiding every r_i. When it exists, the chiral quotients in Theorem 1 arise precisely from the subgroups $N \lhd H$ that are not normal in W and meet the geometric interface. Their conjugacy classes in W have two members, N and r_0Nr_0. The kernel of the flag action of W is*

$$C = \mathrm{Core}_W(N) = N \cap r_0Nr_0.$$

In H/C, the images of these two normal subgroups commute and intersect trivially. In particular, if the flag action is faithful their product is an internal direct product.

Proof. Every such index-two subgroup is the kernel of the homomorphism to C_2 sending each r_i to its nonidentity element. A map on a generating set determines a homomorphism uniquely. This also gives an effective existence test: assign parity to vertices of the Cayley graph, and

reject if any edge contradicts the assignment. Equivalently, the subgroup generated by the $r_0 r_i$ has index two; otherwise no such H exists. Indeed every even word is a product of pairs, and $r_i r_j = (r_0 r_i)^{-1}(r_0 r_j)$.

If $N \lhd H$ is not normal in W, its normalizer contains H but is not W, and therefore equals H. Conversely the normalizer condition of Theorem 1 implies exactly these requirements. Since $W = H \sqcup Hr_0$, conjugation produces just the two displayed subgroups. The usual transitive coset-action calculation gives its kernel as the intersection of all conjugates, hence the displayed C.

Both N and $r_0 N r_0$ are normal in H. The commutator of an element of one with an element of the other belongs to both, hence to C. Their images in H/C therefore commute and have trivial intersection; the product map from their direct product is an isomorphism onto its image. Finally, coset graphs of subgroups $N, M \leq W$ are colour-isomorphic exactly when N, M are conjugate: the same argument as above, for a map $N \mapsto Ma$, gives $aNa^{-1} = M$. Thus the two conjugates describe one unrooted quotient, rather than two different polytopes. □

The condition $C = 1$ is exactly faithfulness of this flag action. We do not infer that an arbitrary monodromy quotient is itself a regular polytope, or identify all notions of minimal polytopal cover. The source's qualified cover statement must be read with this distinction. Its small-cover algorithm removes the conjugate pair just proved. The almost-simple atlas instead reports both enantiomorphic markings before dividing by two; the acquired prefix-coordinate theorem treats those as oriented and unrooted equivalence, respectively.

A literal finite instance. The following instance exercises the interface without assuming any target atlas count. Over $\mathbf{F}_7$, put $V = \mathbf{F}_7^2$, $R = \mathrm{diag}(3, 5)$, and let S exchange the two coordinates. Then R has order six and $SRS = R^{-1}$. In $W = V \rtimes \langle R, S \rangle$, of order 588, use

$$r_0 = ((1, 1), R^3 S), \qquad r_1 = (0, RS), \qquad r_2 = (0, S).$$

These are involutions with pair orders $3, 6, 2$, in string order. They generate: $r_1 r_2 = R$, while $r_0 (R^3 S)^{-1}$ is translation by $(1, 1)$; this vector and its image $(3, 5)$ under R span V. The parity subgroup is $H = V \rtimes \langle R \rangle$.

The only normal subgroups of H that are not normal in W are the two coordinate lines. To see this, if a normal subgroup contains an element with rotation part $R^k \neq 1$, its commutators with V contain $(R^k - I)V = V$, because both diagonal entries of $R^k - I$ are nonzero. It is then the full inverse image of a subgroup of $H/V = C_6$, and is normal in W, since S acts on that cyclic quotient by inversion. Otherwise it is an R-invariant additive subgroup of V. Such a subgroup is an $\mathbf{F}_7$-subspace. The projections $(R - 5I)/(3 - 5)$ and $(R - 3I)/(5 - 3)$ show that it is a sum of coordinate lines. Of the four possibilities, only the two lines are not S-invariant.

The literal checker reconstructs all 588 group elements, verifies the full intersection property of the displayed regular marking, and builds the complete 84-flag quotient for either line. It checks the entire face poset, its flags, diamond intervals and every flag residue, not just the orders of adjacent products. Independently propagating every possible image of one vertex through the coloured graph gives exactly 42 automorphisms, agreeing with H/N. The lines are exchanged by r_0, intersect trivially, and yield one unrooted chiral quotient of this particular regular cover. Its mirror pair is not counted twice. This is a bounded validation example, not a proof of the source's complete $48 + 8$ or 17114 enumerations.

A product obstruction to chiral quotients

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We use the previously established normalizer description of colour-preserving automorphisms of a connected coset flag graph. The following elementary consequence removes an entire class of regular covers before any subgroup enumeration. No priority claim is made for the observation.

Let a regular polytope have string C-group $W = \langle r_0, \dots, r_{d-1} \rangle$. Suppose a quotient polytope has flags Nw, with adjacency $Nw \mapsto Nwr_i$. The assumption that these are its actual flags is essential.

Lemma 1. *Suppose the colours partition into nonempty sets I, J, every generator in I commutes with every generator in J, and $W = W_I \times W_J$. Then*

$$N = (N \cap W_I) \times (N \cap W_J).$$

Proof. Write $n = ab \in N$ with $a \in W_I$ and $b \in W_J$. The flag $Na = Nb^{-1}$ is reachable from N both by an I-coloured path and by a J-coloured path. The first path preserves every face of rank in J; the second preserves every face of rank in I. Hence this flag contains precisely the faces of the flag N, so it is N itself. Thus $a, b \in N$. Every element of N therefore lies in the product of the two displayed intersections. The reverse inclusion is automatic, and the factors commute and meet trivially. □

Theorem 1. *A regular cover with an adjacent product of order two has no chiral polytope quotient with the stated coset flag action.*

Proof. Suppose $r_j r_{j+1}$ has order two. Since the generators are involutions, they commute. The string relations imply that every generator in $I = \{0, \dots, j\}$ commutes with every generator in $J = \{j+1, \dots, d-1\}$. The intersection property gives $W_I \cap W_J = 1$, and generation gives $W = W_I \times W_J$. By the lemma, $N = N_I \times N_J$, where $N_I = N \cap W_I$ and $N_J = N \cap W_J$. Consequently

$$N_W(N) = N_{W_I}(N_I) \times N_{W_J}(N_J).$$

If the quotient were chiral, its automorphism group would have exactly two flag orbits, and every adjacency would exchange them. Any product $r_i r_k$ of two generators therefore returns the base flag to its automorphism orbit, so $r_i r_k \in N_W(N)$ by the normalizer theorem. Take $i \in I$ and $k \in J$. The displayed product decomposition of the normalizer implies $r_i \in N_{W_I}(N_I)$ and $r_k \in N_{W_J}(N_J)$. Varying i and k shows that all generators normalize N. Thus $N_W(N) = W$, and the quotient has one flag orbit, a contradiction. □

The same proof applies to any partition satisfying the lemma, whether or not it arises from a single break in a string. It does not assert that an arbitrary subgroup of a direct product splits; the flag interpretation is what forces that conclusion. Nor does it identify an arbitrary orbit poset with its coset graph.

For ranks one and two there is no separate chiral case. A connected rank-two flag graph is an alternating cycle; rotations by two edges and a reflection preserving colours act transitively on its vertices. Rank one has a single edge. Thus the remaining small-cover search may restrict to rank at least three and to Schlaefli entries all at least three, while still checking the polytopal quotient condition.

This argument uses the source's regular-cover framework and the previous normalizer proof. It is a general exclusion, not a completed atlas census or a solution of a new open question. Completeness of the remaining finite cover roster remains an explicit input obligation.

Chiral quotients with small regular covers

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Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Mathematica Contemporanea 5 (2012), 371–382, report 56 chiral polytopes whose minimal regular covers have at most 2000 flags, excluding covers with 1024 or 1536 flags. We verify the 56 listed examples using the literal companion data acquired on 15 October 2026. This note proves their existence and distinctness. Exhaustiveness among all covers is a separate obligation and is not asserted here. A distinct dual is counted separately, while the two enantiomorphic markings describe one unrooted polytope.

For each entry the retained data specify permutations $r_0, \dots, r_{d-1}$ generating W, and words generating a subgroup $N \leq W$. Put $H = \langle r_0 r_i : 1 \leq i < d \rangle$. The vertices of the flag graph are the right cosets Nw; the edge of colour i sends Nw to Nwr_i. These operations are unambiguous. We acquire the previously proved normalizer theorem: the group of automorphisms preserving every colour is $N_W(N)/N$. In particular, if $N_W(N) = H$ has index two and avoids every r_i, the graph has two automorphism orbits, and each adjacency exchanges them. To obtain a polytope one must also verify the incidence axioms; normalizer data alone do not imply them.

Lemma 1 (Reconstruction from a finite flag graph)**.** *Let a finite graph have d specified involutory adjacencies, none of which fixes a vertex. For each i, call the connected components using all colours except i its faces of rank i. Associate to each graph vertex its tuple of faces. Suppose the following finite checks hold: these tuples are distinct; incidence defined by occurrence in a tuple is transitive; its maximal chains are exactly the tuples; every section of rank two has exactly two intermediate faces; and the graph vertices containing any fixed chain are connected by the remaining colours. Then adjoining a least and a greatest face gives an abstract polytope with precisely the specified flag graph.*

Proof. Ranks orient incidence between distinct ranks. Transitivity, together with equality within a rank, gives a partial order. Each face occurs in a tuple, and each comparable pair does so by definition. Every chain extends in the finite order to a maximal chain, which by hypothesis is a full tuple. Thus all maximal chains have the required length. The section checks are exactly the diamond axiom, including sections incident with the least or greatest face. The connectedness hypothesis is strong flag connectivity. An edge of colour i preserves every other component, and its endpoints differ in the remaining face by injectivity of tuples. The diamond axiom makes this the unique adjacent flag of that colour. Hence the given graph is exactly the flag graph. □

Lemma 2 (Complete comparison of coloured graphs)**.** *Starting at a chosen vertex, traverse a connected graph in breadth-first order, considering colours in a fixed order and numbering each vertex when it is first encountered. The resulting numbered adjacency table is a complete invariant of the graph with its chosen root. Trying each possible image of a root therefore finds all colour-preserving isomorphisms. The least rooted table is a complete unrooted invariant.*

Proof. A rooted colour-preserving isomorphism must carry the first traversal to the second, step by step, so it preserves all assigned numbers and the resulting table. Conversely, equality of tables gives the explicit bijection between equally numbered vertices, preserving every edge. Connectedness makes an isomorphism unique once its root image is fixed. Taking every possible root proves both assertions. □

Theorem 1 (The listed examples)**.** *The retained 56 entries give 48 pairwise nonisomorphic chiral polytopes of rank three and eight of rank four. Each has its stated face numbers, regular monodromy cover and rotation-group SmallGroup identifier.*

Proof. The literal permutation groups have the source orders and SmallGroup identifiers. For every pair of subsets of generators the checker verifies the complete intersection property. It also checks involutions, string commutation and each adjacent product order. Thus these are the stated regular polytopes, using the retained string C-group correspondence.

The stated words generate subgroups of exactly the stated orders. In each case $N \leq H$, $[W : H] = 2$, $N_W(N) = H$, and $\mathrm{Core}_W(N) = 1$. The quotient H/N has the stated SmallGroup identifier, verified on that actual quotient. The core test makes the flag action faithful, so its monodromy group is the literal group W. The normalizer theorem identifies the full automorphism group and its two flag orbits.

For all 19,712 cosets the checker constructs the complete adjacency table. It computes every face component, verifies incidence transitivity, enumerates every maximal chain and checks all diamonds. For each subset of ranks it groups flags by their common faces and compares the resulting sets with components in the complementary colours. Thus every chain residue is connected, and the reconstruction lemma applies. The computed face counts agree with every source entry.

Finally the second lemma checks every possible root image directly from the graphs, without using the group normalizer calculation. For each graph the images of a fixed root under its automorphisms are exactly the even vertices in its bipartition. Their number equals the stated rotation-group order. The 56 least adjacency tables are distinct. This proves chirality, full automorphism orders and the stated equivalence convention independently at the graph level. □

The rank-four rows. The four entries below and their reversals are the eight rank-four examples. We write (n, j) for SmallGroup(n, j); a group order alone is never used to identify a group.

Type	Regular cover group	Rotation group	$\lvert N\rvert$
$\{3, 3, 8\}$	$(768, 1086052)$	$(192, 1494)$	2
$\{3, 6, 9\}$	$(972, 108)$	$(162, 19)$	3
$\{3, 6, 18\}$	$(1944, 949)$	$(324, 77)$	3
$\{6, 6, 9\}$	$(1944, 949)$	$(324, 77)$	3

Certificates, attribution and remaining scope. The pages are parsed as literal integers, disjoint cycles and group words; no downloaded program is executed. Exact source bytes, generated calls and terminal receipts are retained. GAP's finite permutation-group, normalizer, coset-action and SmallGroup identification algorithms remain explicit software premises. The incidence and isomorphism checks use only the complete finite adjacency tables and ordinary integer operations. The rotation relators displayed on the pages are also satisfied, but this does not prove that every displayed abstract presentation has no additional elements. Presentation equality is not claimed by that test.

The source authors retain credit for the examples and classification. The normalizer argument, incidence reconstruction and deterministic rooted comparison supply a uniform verification. No new open-question solution or compression measurement is asserted. The earlier integral proof for the order-588 example is acquired separately. Present website bytes are not asserted to be identical to the 2012 website. A complete reconstruction of the source still needs exclusion of all other regular covers, and its separate almost-simple atlas.

The six group descriptions in the small chiral atlas

Human Compression project

Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Mathematica Contemporanea 5 (2012), 371–382, Table 1 on p. 376, give four rank-four rows. The previously accepted literal-entry component proves the corresponding eight polytopes, including their duals, and identifies their actual cover groups and rotation quotients. Here we verify the structural names, without a census or an inference from group orders or SmallGroup IDs.

We write C_m for the cyclic group of order m and $E_{27} = C_3^3$. The source's D_9 denotes the dihedral group of order 18. Semidirect products assert the existence of the indicated normal factor and complement; their actions need not be specified uniquely by the names. Every parenthesis below is the parenthesis printed in the source.

Lemma 1 (Finite transport and decomposition)**.** *Let $X = \langle x_1, \dots, x_s \rangle$ and $Y = \langle y_1, \dots, y_s \rangle$ be finite permutation groups. Suppose a map $f : X \to Y$ is defined on every element, sends identity to identity, and satisfies $f(xx_i) = f(x)y_i$ for every x, i. Then f is a surjective homomorphism. If $|X| = |Y|$, it is an isomorphism. If instead its identity fibre is N, it induces $X/N \cong Y$.*

If $N, K \le H$, $N \trianglelefteq H$, $N \cap K = 1$ and $|N||K| = |H|$, then $H \cong N \rtimes K$. If also $[N, K] = 1$, then $H \cong N \times K$.

Proof. Following any word in the generators through the edge identities gives the corresponding word in their images. Every group element has such a word, because inverses are positive powers in a finite group. Appending the word for z to the word for x proves $f(xz) = f(x)f(z)$. The images generate Y. The two conclusions follow from the kernel and order formulas. For the second statement, the normal product NK has order $|N||K|$, hence equals H. Multiplication identifies the corresponding semidirect product with H; commuting factors give the direct product. □

The reader constructs complete generated sets by breadth-first closure. It propagates each proposed map from identity and tests every generator edge, rejecting inconsistent assignments. For a proposed factorization, it checks actual membership, conjugation of each normal-factor generator by every ambient generator, trivial intersection and the exact product order. In the direct-product case it checks commutation of the two generator sets. Thus the lemma applies to the literal subgroups, recursively, rather than merely to their names.

The elementary atoms require no naming database. A cyclic generator is checked by closure. Commuting generators of exponent three and order 27 give E_{27}. The dihedral atom is generated by x, y with $|x| = 9$, $y^2 = 1$, $y^{-1}xy = x^{-1}$ and group order 18. For Q_8, the corresponding tests are $|x| = 4$, $y^2 = x^2$, $y^{-1}xy = x^{-1}$ and group order eight. These relations reduce words to the usual distinct normal forms. Finally an explicit generator-edge isomorphism from the full group of 24 permutations on four letters certifies each S_4 atom.

Theorem 1 (The structural assertions of Table 1)**.** *For the retained rank-four entries, the cover and rotation groups have the following structures. The same conclusions hold for their duals.*

Type	*Cover group*	*Rotation group*
$\{3, 3, 8\}$	$((Q_8 \times C_2) \rtimes C_2) \rtimes S_4$	$Q_8 \rtimes S_4$
$\{3, 6, 9\}$	$E_{27} \rtimes (D_9 \times C_2)$	$(E_{27} \rtimes C_3) \rtimes C_2$
$\{3, 6, 18\}$	$(E_{27} \rtimes (D_9 \times C_2)) \times C_2$	$((E_{27} \rtimes C_3) \rtimes C_2) \times C_2$
$\{6, 6, 9\}$	$(E_{27} \rtimes (D_9 \times C_2)) \times C_2$	$((E_{27} \rtimes C_3) \rtimes C_2) \times C_2$

Proof. For each of the eight accepted entries let $W = \langle r_0, r_1, r_2, r_3 \rangle$ be its literal cover group, $H = \langle r_0 r_1, r_0 r_2, r_0 r_3 \rangle$, and let N be generated by the source words. The checker reconstructs these actual permutations. It verifies a full injective generator-edge map from W to its compact certificate model. It separately propagates the images of the three displayed generators of H and checks that the identity fibre is exactly the literal subgroup N. This identifies the rotation quotient with its compact model, including the actual kernel. It also verifies $[W : H] = 2$ and the stated cover and quotient orders.

The 16 atomic and compound registry models have explicit normal factors, complements and generator-edge transports. Applying the lemma at every registry node proves the six displayed names. All eight actual cover and quotient pairs are transported to these same named models. In particular the last two source rows and their duals use identical registry models; their common names are proved by maps, not by repeated order comparisons. The finite verification has 53 maps and 133,673 generator-edge checks. The actual kernel orders are respectively $2, 3, 3, 3$ for the four source rows. □

Attribution and scope. The examples and structural assertions belong to the source authors. The finite transport/decomposition machinery reuses the accepted atlas method; this application supplies explicit certificates and the exact source joins. GAP proposes witnesses only. The final reader uses finite integer permutations and does not trust GAP's proposed isomorphisms, normal subgroups, complements or SmallGroup identifications. It acquires the earlier proof that these literal W, H, N give the stated polytopes. The original failed output-format attempt is retained separately. No exclusion of other covers, complete chiral census, almost-simple atlas result or whole-source completion is asserted here.

Exclusions using the marked generators of a regular cover

Human Compression project

We acquire the normalizer description of chiral quotients in Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Theorem 3.1. We keep every possible flag stabilizer, including those with nontrivial core. The cover action is not assumed faithful. Enumeration of normal subgroups remains an explicit software premise. We also use a necessary condition already implicit in the reconstruction of faces from a flag graph.

Proposition 1. *Let a regular polytope with group $W = \langle r_0, \dots, r_{d-1} \rangle$ cover a chiral abstract polytope, and let N stabilize its base flag. Put*

$$H = \langle r_0 r_1, \dots, r_0 r_{d-1} \rangle, \qquad W_i = \langle r_j : j \neq i \rangle.$$

Then $[W : H] = 2$, $N_W(N) = H$, and

$$N \trianglelefteq H, \qquad N \neq 1, \qquad N \not\trianglelefteq W.$$

The kernel of the cover action is $N \cap N^{r_0}$, which may be nontrivial. Moreover $\bigcap_{i=0}^{d-1} NW_i = N$.

Proof. The subgroup generated by the displayed pairs contains every product of two generators, since

$$r_i r_j = (r_0 r_i)^{-1} (r_0 r_j), \qquad r_0 r_0 = 1.$$

It consists of the even words and has index at most two. In a chiral flag graph every adjacency interchanges the two automorphism orbits. An odd word cannot stabilize a flag. The even subgroup therefore has index two and agrees with the normalizer of N, by the acquired normalizer theorem.

The normal-subgroup conditions follow from the same theorem. Since N is normal in the subgroup H of index two, its only conjugates in W are N, N^{r_0}. Their intersection is therefore the core and the kernel of the coset action. If $N = 1$, its normalizer would be all of W, contrary to chirality.

Identify graph vertices with right cosets Nw, with colour j sending Nw to Nwr_j. The face of rank i through the base flag is precisely the connected component obtained by omitting colour i. Its vertices are the cosets Nh with $h \in W_i$. Hence Nw has every base face if and only if $w \in NW_i$ for every i. A flag is determined by its faces; such a vertex must equal N, so $w \in N$. The reverse inclusion is immediate. □

For each literal source marking, the reader checks the generated group, its order, the involutions, adjacent orders and string commutations. The database identifier is a source label; no abstract identification is used by this test. If $H = W$, this marking is excluded. Otherwise it enumerates all normal subgroups of H and retains exactly the candidates satisfying the displayed group conditions. For each it computes all NW_i. If T_i is a right transversal of $N \cap W_i$ in W_i, then $NW_i = \coprod_{t \in T_i} Nt$; this is the finite computation used. A negative result also retains a literal element $w \notin N$ and checks $n_i^{-1} w \in W_i$ for some $n_i \in N$ at every rank.

Passing this test does not prove that the quotient is a polytope. Survivors still require the full face and flag axioms and comparison up to coloured graph isomorphism. No new exclusion is credited before the individual source entry has finished and been reviewed. The earlier filter for faithful actions alone does not exclude arbitrary cover actions. The source examples and original normalizer enumeration retain their original attribution. This component supplies an explicit finite application of the acquired group and face criteria, not a new complete atlas theorem.

Identifying the quotient flag graphs

Human Compression project

This note completes the finite calculation for the first atlas in Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Math. Contemp. 5 (2012), 371–382. The second atlas, for almost simple automorphism groups, is a separate outstanding calculation. We acquire the earlier complete atlas of regular covers, the reviewed normalizer criterion and marked-cover obstruction, and the explicit verification of the 56 positive examples. In particular, the completeness of the earlier regular atlas is a retained input. Current website files are bound literally; historical byte identity is not asserted.

Let $W = \langle r_0, \ldots, r_{d-1} \rangle$ be a supplied marked permutation group and let $N \leq W$ be one of its retained flag stabilizers. All cores are allowed. The first calculation supplies $|W|$, the generators of N, its order, and the conditions on its normalizer. Its subgroup enumeration uses the stated GAP software premises. The following additional check uses only literal permutations and finite graphs.

Proposition 1 (Checking a coset graph)**.** *Let X be a connected graph with one adjacency $x \mapsto x^i$ of each colour i at each vertex, and choose $x_0 \in X$. Choose permutations $g_x \in W$ by following a breadth-first traversal from $g_{x_0} = 1$. Suppose that*

$$|X|\,|N| = |W|, \qquad g_x r_i g_{x^i}^{-1} \in N \quad (x \in X,\ 0 \leq i < d).$$

Then $x \mapsto Ng_x$ is an isomorphism from the coloured graph X to the right coset graph of N in W with the given generators.

Proof. The displayed membership is exactly the equality $Ng_x r_i = Ng_{x^i}$. Thus the map preserves every adjacency and sends x_0 to N. Every element of W is a word in the marked generators. Following that word from x_0 shows that every coset occurs in the image. The cardinality equality makes this surjective map bijective. □

The checker enumerates N from its literal generators and verifies its claimed order. It constructs every g_x as a word in the original marks and checks every displayed equality. The value of $|W|$ is acquired from the preceding exact group calculation. An abstract database identifier is not used in this argument.

Proposition 2 (Reusing a verified flag graph)**.** *In a connected graph with specified colours and one adjacency of each colour, an isomorphism is determined by the image of one vertex. Label the vertices in breadth-first order, always exploring colours in their specified order, and record the resulting adjacency array. Two rooted graphs have the same array if and only if the traversal orders give a colour-preserving isomorphism between them.*

Proof. An isomorphism must send each path from the root to the path with the same colours from its image. Consequently it preserves the traversal order and recorded array. Conversely equality of the arrays makes the bijection between the two traversal orders preserve every coloured edge. □

Each of the 56 acquired chiral flag graphs has exactly two automorphism orbits, represented by a chosen vertex and one of its adjacent vertices. The 112 corresponding arrays therefore suffice for comparison with every possible root image. Every successful comparison also supplies

the full vertex bijection, and every edge of that bijection is checked. The acquired polytope axioms, chirality and group identifications then transfer through this explicit isomorphism.

For an unmatched graph, omit colour i to obtain its prospective rank-i faces as connected components. In an actual polytope these are precisely the faces containing the flags in those components. Two particular necessary conditions suffice for all unmatched graphs in this calculation. First, each interval between ranks $i-1$ and $i+1$ has exactly two intermediate faces, with the least and greatest faces included at the boundary ranks. Second, flags containing the same fixed chain are connected by adjacencies that change none of those faces. The first condition is the diamond axiom; the second is strong flag connectedness. A failure is recorded by the actual face labels and representative flags, or by two flags in distinct allowed components with the same fixed faces. These are necessary conditions in every rank; passing them alone is not used as a positive certificate.

The complete calculation. The current regular-atlas input contains 9,212 nondegenerate markings. The earlier product argument excludes the markings with a bond labelled two; ranks one and two cannot be chiral. The three retained execution intervals cover respectively the source indices $[0, 118)$, $[118, 7302)$ and $[7302, 9212)$, without overlap. The first two interval receipts remain failed or interrupted; their individual completed entries are acquired from the exact completion journal and saved records. No interrupted entry contributes an exclusion.

The normalizer conditions yield 8,986 candidates across those markings. The base-face intersection test leaves 756. Literal coset transport checks 704,608 coloured edges of these quotient graphs. Of the 756, 360 have explicit isomorphisms to the acquired examples and cover all 56 of those examples. The other 396 have recorded obstructions: 350 violate a diamond condition and 46 violate flag-residue connectivity. There are no undecided candidates. The verified examples are pairwise nonisomorphic as coloured flag graphs, so this gives exactly 48 of rank three and eight of rank four, with a distinct dual counted separately.

Scope and attribution. The result is relative to the stated complete regular-cover input and exact group-arithmetic premises. The normalizer description and the original 56 examples retain their original attribution. The contribution here is a complete calculation with arbitrary action kernels and explicit transport to reusable verified examples, together with literal negative witnesses. No claim about the second atlas, a new open question, Lean formalization or measured compression follows from this component.

Counting chiral markings by stabilizers

Human Compression project

The second atlas in Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups* (2012), specifies many finite automorphism groups. We apply the already reviewed prefix characterization and enumeration theorem to each group. This note supplies an orbit count and its concrete implementation. No numerical entry of the second atlas is assumed as an input.

For a finite group G, an admissible prefix is an ordered tuple $(1, a_1, \dots, a_{n-1})$ satisfying the acquired adjacent-order, nonadjacent-involution and difference-subgroup intersection conditions. For $J \subseteq \{0, \dots, n-1\}$ write

$$D_J = \langle a_i^{-1} a_j : i, j \in J \rangle.$$

The shared coordinate theorem identifies these with the corresponding interval-rotation subgroups. The shared search theorem proves that prefixes remain admissible, each extension strictly enlarges the generated subgroup, and an admissible generating tuple cannot be extended further. The Schulte–Weiss reconstruction theorem remains the attributed geometric input.

Proposition 1. *Choose one representative a_1 from each conjugacy class of elements of order greater than two in G. At a proper prefix generating H, choose one representative of each $C_G(H)$-orbit on the permissible next involutions, and retain precisely the extensions passing the intersection tests. Stop a branch when its entries generate G. The generating tuples so obtained are representatives of exactly all inner automorphism orbits of admissible generating tuples.*

Proof. Conjugation preserves every admissibility condition. After the first entry has been moved to its chosen representative, precisely its centralizer remains available to move the next entry. More generally, an element fixes every entry of a prefix if and only if it centralizes the subgroup they generate. The permissible next entries form an invariant set under this centralizer. Choosing one representative of each such orbit therefore gives exactly one representative of each orbit of the extended prefixes over the fixed prefix. Induction proves both completeness and uniqueness of the orbit representatives at every depth. The acquired hereditary and strict-growth statements justify the retention tests and stopping rule in every possible rank. □

The implementation first filters the involutions by the order conditions and by exclusion from H. These tests are invariant under $C_G(H)$, so they can precede the orbit computation. The intersection property is also invariant; it is tested on one representative per orbit. Every difference subgroup is computed from a star at its least index, by the shared star-generation identity. The checker tests all pairs of subsets, including empty intersections. It does not substitute a necessary condition for the complete intersection property.

Proposition 2. *Let $\mathcal{T}$ be a finite, $\mathrm{Aut}(G)$-invariant set of ordered generating tuples. If b is the number of its inner automorphism orbits, then the number of its full automorphism orbits is*

$$\frac{b}{|\operatorname{Out}(G)|}, \qquad |\operatorname{Out}(G)| = \frac{|\operatorname{Aut}(G)|\,|Z(G)|}{|G|}.$$

Proof. An automorphism fixing all entries of a generating tuple fixes every element of G. Thus $\mathrm{Aut}(G)$ acts freely on $\mathcal{T}$, and so does its subgroup $\mathrm{Inn}(G)$. Each full orbit has $|\operatorname{Aut}(G)|$ elements

and each inner orbit has $|\operatorname{Inn}(G)| = |G|/|Z(G)|$ elements. A full orbit is a union of precisely $|\operatorname{Out}(G)|$ inner orbits. Dividing the total number of inner orbits by this common number proves the claim. □

Rank, the ordered Schlaefli type, and the existence of a reflection automorphism are invariant under $\operatorname{Aut}(G)$. The proposition therefore applies separately in each such class. The reflection test is the acquired criterion that an automorphism send every a_i to a_i^{-1}. A homomorphism with these images, if it exists, is surjective because the images generate G; finiteness makes it an automorphism. We use the checked homomorphism constructor, not its unchecked variant. An unsuccessful constructor excludes the reflection relative to the stated exact group-arithmetic premise.

Consequently the number of chiral oriented classes of a given rank and type is its number of nonreflectable inner representatives divided by $|\operatorname{Out}(G)|$. The acquired mirror theorem pairs these classes freely when orientation is forgotten. Duality is still kept distinct. The regular tuples in this calculation concern rotation groups; they are not identified with the source's separate column counting regular polytopes by their full automorphism groups.

Corollary 3. *A finite chiral polytope cannot have an adjacent Schlaefli entry two.*

Proof. The corresponding flag adjacencies commute. Together with the distant commuting relations, this partitions the colours into two nonempty sets I, J whose generated permutation groups M_I, M_J commute. Let $W = M_I \times M_J$ act on the actual flags through multiplication of these commuting permutations. It acts transitively, so the flags are $N\backslash W$ for the stabilizer N of a chosen flag. The previously proved flag-component argument gives $N = (N \cap M_I) \times (N \cap M_J)$: if $ab \in N$, the flag reached by a is also reached by b^{-1}, and these two paths preserve complementary sets of faces, hence preserve the whole base flag. The normalizer splits as a product as well. Chirality would put every even word $r_i r_j$ in this normalizer. Taking $i \in I, j \in J$ then puts both individual generators in their respective normalizer factors. All generators normalize N, making the polytope regular, a contradiction. This uses the actual flag action; no unproved intersection property is imposed on the monodromy group. □

Verification boundary and attribution. The coordinate and prefix theory, reflection criterion and geometric reconstruction are acquired from the reviewed manuscript on the algorithm of Buekenhout, Leemans and Tranchida, with the attributions retained there. The centralizer recursion uses the same orbit argument as other parts of the project. The count of free orbits is elementary; no priority claim is made. The last corollary extends the previously reviewed product proof by using the direct product of the two actual monodromy subgroups.

The program retains GAP's exact group operations, including complete conjugacy classes, centralizers, orbits, subgroup intersections, automorphism groups and homomorphism construction, as software premises. Literal small-group controls compare its complete output with the independently acquired Cayley-table search. Large finite target calls require their own complete receipts, input identities and distinct review; this note alone confers no source census or whole-paper credit.

Partitioning the first conjugacy classes

Human Compression project

We acquire the reviewed prefix enumeration and its exact implementation. For a fixed literal permutation group G, let $(a_1, \ldots, a_s)$ be the complete ordered list of conjugacy class representatives of order greater than two used by that implementation. Its outermost loop starts the prefix $(1, a_i)$ once for each i. All later searches use the centralizer of the current prefix and the unchanged intersection tests.

Proposition 1. *Executing those s outermost branches separately and taking the disjoint union of their returned lists gives the same complete collection of inner orbits as the acquired enumeration. This remains true when a completed branch is acquired from an earlier execution, provided its exact group, ordered representative list and implementation are bound.*

Proof. The recursive visit uses only the fixed group, its involutions, the current prefix and its parabolic subgroups. Earlier results and counters are never read in selecting an extension or testing a marking. Restricting the outer loop to one index therefore leaves its complete mathematical output unchanged. An inner automorphism relating two resulting markings must conjugate their first nonidentity entries. Distinct entries of the list represent distinct conjugacy classes, so different branches cannot represent the same inner orbit. Conversely every admissible generating marking has a first nonidentity entry in one of those classes. The acquired completeness theorem places its orbit in that branch. □

Each returned file includes the whole ordered representative list, the chosen index and the exact literal marking arrays. The collector requires identical literal group inputs and representative lists, identical centre and automorphism orders, and exactly the indices $1, \ldots, s$, each once. It checks that every returned marking starts at its branch representative. If a group algorithm chooses different representatives on another run, the collector rejects that join; an index alone is never its identity. No previous interrupted branch supplies a count. A deadline leaves that branch unpaid and does not erase completed files for other branches.

After the complete union, the acquired free-action argument divides the inner count in each rank, ordered type and chirality class by $|\operatorname{Out}(G)|$. This division is not made within an individual first branch: an outer automorphism can exchange first conjugacy classes. The regular entries in this output still mean directly regular rotation groups. They are not the regular columns of the almost-simple source atlas.

The recursion, group arithmetic and full-automorphism computation retain their earlier proofs and software premises. The present change is only the finite partition and exact receipt union. The writer uses a wider GAP output line to avoid adding line-continuation escapes to JSON fields. No target count, source admission, new conjecture or formalization follows from the method alone.

A fixed first-class plan for the prefix counter

Human Compression project

We acquire the accepted prefix enumeration, first-class partition proof, and version-two receipt admission. Different executions of the group library may return different ordered representatives of the same complete conjugacy classes. Version two correctly rejects their literal union. The present caller instead takes one explicit, hash-bound list and validates it against the actual group on every invocation. No random seed or equality of class indices is assumed.

Let G be the fixed literal finite permutation group. A proposed plan is an ordered list $(a_1, \dots, a_s)$ of permutations on its actual moved-point domain. Independently compute the complete conjugacy classes of G, and let $(b_1, \dots, b_t)$ be their representatives of order greater than two. The completeness of this library interface is the same retained software premise as in the accepted counter.

Proposition 1. *Suppose $s = t$, every a_i belongs to G and has order greater than two, and there is a permutation π of $\{1, \dots, s\}$ and elements $g_i \in G$ satisfying $a_i^{g_i} = b_{\pi(i)}$. Then the fixed list contains exactly one representative of every conjugacy class of order greater than two. Executing only branch i of the accepted counter from $(1, a_i)$, and joining exactly the branches $1, \dots, s$, therefore enumerates exactly the same inner marking orbits as the accepted whole counter.*

Proof. Surjectivity of π places every required conjugacy class in the fixed list. If two fixed entries were conjugate in G, their corresponding b entries would be conjugate and hence represent the same library class, contradicting injectivity of π. The fixed entries therefore are a complete transversal. The acquired partition proof applies to any such transversal: every marking has its first entry in exactly one of these classes, while subsequent prefix centralizers and intersections are computed by the unchanged recursion. Conjugating a marking transports it to the selected representative. Consequently different first branches cannot duplicate an inner orbit and every orbit occurs. Earlier results and counters are not read when choosing a recursive extension. □

The helper checks actual membership and order. For each fixed entry it finds an actual conjugating element to one fresh representative, checks that element's membership in G and the conjugacy equality, and rejects any reused class. Equal list lengths and the checked bijection prove coverage. The returned certificate stores the two full literal lists, the bijection and all conjugating permutations. The Python admission reader independently verifies the permutation domains, bijection and literal conjugacy identities; group membership and fresh-class completeness are acquired from the exact executed helper. A proposal extracted from an older census file is only a proposal: its prior count or claimed class indices confer no validity.

The only scientific code changes are this helper, two added arguments, and replacement of the original first-list assignment by the fixed list and its validation. The complete recursive visit, subsequent outer loop, marking tests, automorphism computation and result writer are byte-identical to version two. The full involution list still comes from the fresh complete conjugacy classes.

Receipt boundary. Every invocation requires the exact new operational review, with its exported dependency file binding this author edition. All earlier manifest/export/dependency checks remain. The plan and model are mandatory pinned inputs. The request and driver are reconstructed from their exact bytes; the driver loads the actual new counter and embeds the full fixed list. The collector requires the precise command, clean terminal status, complete unchanged

input and output inventories, and the returned plan certificate. Every census must contain the entire pinned list and start at the requested representative. A union requires one identical literal model, plan, engine, author and operational acceptance, and exactly every index once. An external deadline grants no branch credit. Neither a partial class list nor a differently located plan is silently substituted. Division by the outer automorphism index is permitted only after this complete union, exactly as before.

Evidence and scope. The retained version-two L3(4) extension calls returned with differing literal class lists; their rejected union and complete individual files remain historical evidence, not a new union certificate. Small actual A_4 controls compare two complete fixed branches with the acquired whole result. Separate actual group checks reverse and conjugate the backend representatives while keeping the fixed list unchanged, and also reverse the proposed plan. Duplicated, missing, extra, identity, involution and outside-group entries are rejected. Receipt and gate damages preserve the distinction between explicit test-only fixtures and production admission. No source-group branch is executed by these checks.

The original partition theory and counter were authored by the parent agent; this fixed-plan helper and operational successor were authored by the reviewer of the former receipt interface, with distinct review of this new work reserved to the parent. No source count, mathematical classification, whole-source completion or performance improvement is claimed by this caller alone.

A common action for a group and its automorphisms

Human Compression project

The acquired regular-polytope enumerator requires permutation groups $G \triangleleft A$ in the same action, with $|A : C_A(G)| = |\operatorname{Aut}(G)|$. A separately named permutation model of $\operatorname{Aut}(G)$ does not by itself supply this input. We use the following elementary construction to connect already identified models to the acquired enumerator.

Proposition 1. *Let H be a finite permutation group and $a = |\operatorname{Aut}(H)|$. Either of the following constructions supplies a faithful copy G of H and a conjugating group A inducing every automorphism of G.*

1. *Begin with $A = H$ in its given action. Adjoin permutations which normalize H. Accept this action when $|A : C_A(H)| = a$ and put $G = H$.*

2. *Let I be the complete set of involutions of H. Let G be the image of the conjugation action of H on I, and let A be the image of the action of $\operatorname{Aut}(H)$ on I. Accept these actions when both are faithful. Then $G \triangleleft A$ and $|A : C_A(G)| = a$.*

Proof. In the first construction, the supplied permutations normalize H, so the resulting group A does too. Conjugation gives a homomorphism $A \to \operatorname{Aut}(H)$ with kernel $C_A(H)$. Equality of the finite orders implies that its image is all of $\operatorname{Aut}(H)$.

For the second construction, automorphisms preserve element order, so I is invariant under both actions. Write $\iota : H \to \operatorname{Sym}(I)$ and $\alpha : \operatorname{Aut}(H) \to \operatorname{Sym}(I)$ for their homomorphisms. Their images are G and A. Faithfulness of ι identifies H with G; this is a checked condition, not an assumption based on the group name. For $h \in H$ and $\phi \in \operatorname{Aut}(H)$, the conjugation actions satisfy

$$\alpha(\phi)^{-1}\iota(h)\alpha(\phi) = \iota(\phi(h)),$$

with permutations acting on the right. Hence G is normal in A, and the induced automorphisms of G are precisely the transports of all automorphisms of H. If $\alpha(\phi)$ centralizes G, the displayed equality and injectivity of ι give $\phi(h) = h$ for every h. Thus $C_A(G)$ is trivial and $|A : C_A(G)| = a$. □

In the implementation, a proposed normalizing permutation is tested on every generator of H. Inclusion $H^c \leq H$ implies equality because conjugation preserves the finite order. Candidate overgroups are only a way of obtaining a smaller faithful action; their names and orders are never evidence for the automorphism condition. If none suffices, the implementation forms I from all involution conjugacy classes and checks its cardinality against their sizes. The orders of both action images certify faithfulness. It additionally checks normality and the centralizer quotient directly.

The generator images of H under ι retain the connection to the original named model. The acquired enumerator then applies unchanged to (G, A), and an isomorphism preserves generation, string relations, all subset intersections and equivalence under automorphisms. Its ordered classifications therefore transfer to H. Completeness still requires a complete returned enumeration; successful action setup alone is not a polytope census.

The use of involutions as a common set is borrowed from the already reviewed exceptional Coxeter computations in F006. The enumeration, intersection criterion, rank bound and exact GAP operations are acquired from the reviewed L071 application. This component provides the compatibility argument and explicit checked interface, not a new classification theorem or an open-conjecture solution.

The regular columns of the small-group atlas

Human Compression project

We consider the 79 group models printed in Tables 2–7 of Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Mathematica Contemporanea 5 (2012), 371–382. There are 88 table occurrences, since some groups occur more than once. A regular marking of rank r means an ordered tuple of nonidentity involutions $\rho_0, \dots, \rho_{r-1}$ which generates the group, satisfies $\rho_i\rho_j = \rho_j\rho_i$ for $|i-j| > 1$, and has the intersection property

$$\langle \rho_i : i \in I \rangle \cap \langle \rho_j : j \in J \rangle = \langle \rho_k : k \in I \cap J \rangle \qquad (I, J \subseteq \{0, \dots, r-1\}).$$

Markings are identified by automorphisms carrying each generator to the corresponding generator. Reversal is not additionally identified. The standard correspondence with regular abstract polytopes is retained; we do not assert a classification under a different incidence convention.

Proposition 1. *Relative to the acquired literal models, group-operation implementations, and the reviewed enumeration theorem, the regular columns for all 79 printed groups are determined by the accompanying census files. Of the 88 printed rank-count rows, 87 agree with these results. In Table 4, page 379, the rank-three entry for* $\mathrm{PSL}_2(27)$ *is 26, whereas the source prints 27. All other regular entries agree rank by rank.*

Proof. Fourteen models are supplied by the already reviewed complete regular censuses; their rank lists are acquired without repeating their searches. For the remaining 65 models, the acquired enumeration constructs involutory prefixes up to the complete automorphism action. At an extension, representatives of the relevant double cosets enumerate all inequivalent choices in the required centralizers. The intersection test and the acquired normal-closure rejection remove only impossible prefixes. A string C-group tuple is independent: removing a generator strictly reduces the subgroup it generates. The increasing subgroup chain therefore bounds its rank by $\lfloor \log_2 |G| \rfloor$. These are the hypotheses and conclusion of the previously reviewed enumeration theorem; this application uses that theorem without altering its search or rejection rules.

The action on each literal model is acquired from the reviewed action construction. For a normalizing permutation group A, the equality $|A : C_A(G)| = |\operatorname{Aut} G|$ proves that its conjugation action is complete. Where necessary, the construction instead uses the faithful common action of G and its automorphism group on the involutions. The checked generator map connects that action to the original model. Both constructions and their faithfulness hypotheses are retained inputs.

Every one of the 65 actual executions returned its complete rank list, with a clean terminal result and unchanged input files. The independent receipt verifier reconstructs the complete input, driver and output inventories and reparses the returned census. It neither executes another search nor reads the source counts as search inputs. Joining these 65 lists with the 14 retained lists gives exactly the 79 printed group keys, without overlap. Joining them to all 88 source occurrences gives the stated comparison. The machine-readable result records every rank list, including the source discrepancy and all duplicate joins. □

An independent rank-three check for $\mathrm{PSL}_2(27)$**.** This count also has a short finite verification which does not use the prefix enumeration. Write $G = \mathrm{PSL}_2(27)$ in its natural action on 28 points. Its order is 9828. The accompanying script checks literal equality with the standard constructor, and acquires a normalizing action of order 58968 inducing all automorphisms. There are 351 involutions in G, forming one orbit. Fix an involution x. The script lists all 14 distinct involutions commuting with x and verifies that the stabilizer of x is transitive on them. Thus all possible ordered commuting endpoint pairs (x, z) are equivalent. The stabilizer of one such pair has order 12.

For this fixed pair, the middle generator y must be one of the 351 involutions. Exactly 312 choices generate G together with x, z. For every one of them the explicit subgroup computation gives

$$\langle x, y \rangle \cap \langle y, z \rangle = \langle y \rangle.$$

This is the rank-three intersection criterion: the rank-two subgroups are dihedral string C-groups, and their intersection is the common rank-one subgroup. Equivalently, the other rank-two intersections follow from $\langle x, z\rangle \cong C_2^2$, the displayed equality, and generation. The endpoint stabilizer acts freely on the 312 possibilities, because an automorphism fixing the three generating elements is the identity. Consequently the number of rank-three markings is $312/12 = 26$. The script separately checks its orbits and their sizes. Enumeration of all elements, subgroup operations and the automorphism-group order are explicit GAP premises of this independent finite check. This paragraph makes no assertion about higher ranks on its own.

Scope and attribution. The source determines the group roster and the marking convention. The previous complete regular censuses, the enumeration theorem and the common-action construction supply the reused mathematics. The new work here is their complete application to the 65 remaining models, the join to all printed regular entries, and the separate fixed-endpoint check of the exceptional row. This is a correction of a finite published count, not a claimed solution of an open conjecture or a claim of priority. The chiral columns, first atlas, and the scope of the historical ATLAS roster require their separate arguments. No whole-manuscript completion, Lean verification or compression measurement follows from this component.

Literal chiral markings in $\mathrm{PSU}_4(2)$

Human Compression project

We prove a positive lower bound using explicit permutations and relation words. The finite search which found them is not a premise of this lower bound. The named group is the acquired degree-forty model of $\mathrm{PSp}_4(3) \cong \mathrm{PSU}_4(2)$, of order 25920; its identification is retained from the reviewed subgroup atlas. In particular, it is not identified by its order alone.

We use the already acquired geometric recognition theorem of Schulte and Weiss, in the prefix coordinates of our reviewed manuscript. At rank three the tuple $(1, s, t)$ represents a chiral marking when

$$\langle s, t\rangle = G, \quad |t| = 2, \quad |s|, |s^{-1}t| > 2,$$

the three cyclic subgroups $\langle s\rangle$, $\langle t\rangle$, $\langle s^{-1}t\rangle$ intersect pairwise trivially, and no automorphism simultaneously inverts s and t. The cyclic subgroups are precisely the three nontrivial proper difference subgroups. The remaining difference subgroups are 1 and G, so these tests imply the full intersection property. We count ordered markings modulo $\mathrm{Aut}(G)$, retaining the ordered type and distinguishing mirror markings. This convention is explicit throughout.

Lemma 1. *Suppose s, t and s', t' each generate the same finite permutation group G. Start with the proposed image $1 \mapsto 1$, and propagate images by the directed edges $x \mapsto xs$ and $x \mapsto xt$, using target edges $y \mapsto ys'$ and $y \mapsto yt'$. If propagation is consistent on every edge, it defines the unique automorphism sending (s, t) to (s', t'). If it is inconsistent, there is an explicit word w with $w(s, t) = 1$ and $w(s', t') \neq 1$.*

Proof. Every element is a positive word in s, t, since inverses are positive powers in a finite group. The directed graph is therefore connected from the identity. Consistency means that the proposed image is independent of the chosen positive word: follow the letters of any such word along the verified edges. Consequently concatenating words shows that the resulting function preserves multiplication. It is surjective because s', t' generate G, hence bijective by finiteness. Its generator images make it unique.

For an inconsistent edge, let u be the recorded word reaching its initial vertex and let v be the recorded word reaching its final vertex. If the edge is labelled $z \in \{s, t\}$, the word uzv^{-1} evaluates to the identity in the source. The two conflicting target images show that it does not evaluate to the identity in the target. This gives the claimed obstruction, which can be checked by permutation multiplication without repeating the propagation. □

Proposition 2. *The supplied permutations give at least 46 distinct ordered chiral marking classes of rank three in* $\mathrm{PSU}_4(2)$*. Their ordered types and multiplicities are as follows.*

Type	Count	Type	Count	Type	Count	Type	Count
$(4, 9)$	2	$(5, 9)$	6	$(9, 4)$	2	$(12, 4)$	2
$(4, 12)$	2	$(5, 12)$	2	$(9, 5)$	6	$(12, 5)$	2
$(6, 9)$	2	$(9, 6)$	2	$(9, 12)$	4	$(12, 9)$	4
$(6, 12)$	2	$(9, 9)$	4	$(12, 6)$	2	$(12, 12)$	2

Proof. The certificate contains 92 literal tuples $(1, s, t)$ on forty points. The validator first closes the acquired ambient generators under permutation multiplication and obtains 25920 elements. For each tuple it verifies membership, the displayed order conditions, the three cyclic intersections

and equality of its generated subgroup with the ambient group. These are finite set equalities on actual permutations.

Apply Lemma 1 with target (s^{-1}, t^{-1}) to each tuple. Every propagation yields a relation obstruction. The retained 92 words therefore exclude every required reflection. The acquired recognition theorem now supplies a chiral polytope marking in each case.

To distinguish the markings, process them in their retained order. Compare a tuple with the existing class representatives of its same ordered type using Lemma 1; different types cannot be related by an automorphism. A successful propagation assigns the tuple to that representative. If all comparisons fail, retain a new representative. Induction shows that every new representative is inequivalent to all earlier ones, and every assigned tuple is equivalent to its recorded representative. The completed comparisons give 46 classes, each containing two of the supplied tuples. There are 46 successful transports, each checking 25920 vertices and 51840 directed edges, and 118 unsuccessful transports with explicit relation witnesses. The classes have precisely the sixteen distributions in the table. Neither a computed outer automorphism order nor completeness of the search is needed for this argument. □

For example, the first marking has type $(12, 9)$ and is given by the following permutations, with omitted points fixed:

$$\begin{aligned} s = {} & (1,24)(2,26,3,31)(5,12,10) \\ & (6,40,9,15,13,35,7,17,8,33,11,22) \\ & (14,38,37,16,19,36,32,18,21,34,39,20) \\ & (23,25)(27,28,29,30), \\ t = {} & (1,2)(3,4)(5,14)(6,20)(7,17)(8,15)(9,21) \\ & (10,18)(11,16)(12,22)(13,19)(23,32)(24,38) \\ & (25,35)(26,33)(27,39)(28,36)(29,34)(30,40)(31,37). \end{aligned}$$

One reflection obstruction is

$$w(s,t) = s^2ts^5tst^{-1}s^{-1}t^{-1}s^{-3}t^{-1} = 1, \qquad w(s^{-1}, t^{-1}) \neq 1.$$

Both statements are checked by direct multiplication with permutations acting on the right. The complete certificate supplies the same form of witness for every proposed reflection and every rejected comparison.

Source comparison and attribution. Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Mathematica Contemporanea 5 (2012), 371–382, print 26 in the chiral column of the PSU$(4, 2)$ row of Table 6, on page 381. Section 4 classifies the ordered generator tuples up to isomorphism. Section 6 explicitly counts the two enantiomorphic forms separately, and halves its overall total only when these are subsequently identified. Thus the printed column uses the same convention as the proposition: its value 26 is contradicted by the literal lower bound 46. This comparison does not identify duals. The mirror formula in Section 3 becomes (s^{-1}, t) in our prefix coordinates, exactly the reflection tested above. The broader search reports the same count 46, but its exhaustion is not used in this refutation of the printed value.

The geometric reconstruction and named-group identification are borrowed, with the attributions retained in the acquired manuscripts. The present contribution is the explicit finite collection and its verification by permutation closure and relation words. This verifies a positive lower bound in ordinary Python, not a Lean formalization or a new general reconstruction theorem. No compression factor or completion of the whole source follows from this component alone.

The printed almost-simple chiral atlas: complete applications and its domain

Private L070 recovery component

The fixed source is Hartley–Hubard–Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Math. Contemp. 5 (2012), 371–382. We distinguish its 79 printed group models, their 88 occurrences in Tables 2–7, and the larger domain stated in Section 4. All computations below use those exact named models; printed polytope counts are never search filters. The separate first-atlas classification is not repeated here.

Convention and retained recognition. An admissible prefix marking is an ordered tuple

$$\alpha = (1, \alpha_1, \ldots, \alpha_{r-1})$$

generating G, with the adjacent orders greater than two, nonadjacent differences involutions, and the full intersection property for the subgroups generated by $\alpha_i^{-1}\alpha_j$ on each index set. The acquired recognition theorem gives the corresponding regular or chiral abstract polytope. A marking is regular exactly when the simultaneous inversion of $\alpha_1, \ldots, \alpha_{r-1}$ extends to an automorphism. Markings are identified by ordered generator images under $\mathrm{Aut}(G)$; reversal is not additionally identified. In the chiral part the source counts the two enantiomorphic forms separately: Sections 3 and 4 give the ordered-marking construction, and Section 6 expressly divides its count by two only when those forms are to be identified. This convention was independently checked in the acquired source-orientation review.

Theorem 1. *Relative to the acquired named models, recognition and exhaustive-prefix theorems, and the stated finite group-operation implementations, all chiral columns for the 79 printed groups are determined. Their total is*

$$16974 = 15056 \text{ (rank 3)} + 1890 \text{ (rank 4)} + 28 \text{ (rank 5)}.$$

Of the 88 printed rank distributions, 87 agree. Table 6, page 381, prints 26 for $\mathrm{PSU}_4(2)$*; the correct rank-three count in the same convention is 46. There are no other rank-count discrepancies in the printed chiral rows.*

Proof. The accepted prefix theorem enumerates inner-conjugacy classes by choosing one representative of each nonidentity, noninvolutory first conjugacy class, and then representatives under the centralizer of the prefix subgroup at every extension. Its full intersection test and reflection test are unchanged. Every extension strictly increases that subgroup, so the finite enumeration exhausts all ranks, rather than stopping at the largest rank printed in the source.

The actual executions consist of 72 clean complete whole-group calls, one complete 25-class union for S_9, and six complete fixed-plan unions for $L_3(4).D_{12}$, $P\Sigma U_3(5)$, J_2, $L_3(5) : 2$, $P\Gamma U_3(5)$ and $M_{22} : 2$. The last six replace six earlier nonreturns, which receive no census credit. Each fixed-plan call checks its full pinned first-representative list against freshly computed complete classes by actual conjugating transporters. The receipt union requires the exact same literal list, model, engine and acceptance, and each index exactly once. In particular an index from a different randomly ordered class list cannot be substituted.

The application reader reconstructs the whole-call drivers and exact input/output inventories, checks all ordered class-return markers and the terminal census marker, and requires a clean terminal process. For class unions it invokes the unchanged accepted v2 or v3 receipt verifier in isolated Python processes. The fresh application makes no group calls. All literal census files equal their previously acquired unions. The older whole-call receipts bind their exact program, model and driver, but record only the GAP executable locator, not a complete runtime-file trace; that historical provenance limit is retained explicitly. The stronger v2/v3 runtime inventories are not retroactively attributed to them.

Finally $\mathrm{Out}(G)$ acts freely on inner classes of generating markings. Indeed, an automorphism stabilizing an inner class agrees on every generator with an inner automorphism, hence equals it. Regularity, rank and ordered type are invariant. Thus each returned rank/type/reflection stratum has size divisible by $|\mathrm{Out}(G)|$,

and division gives exactly the ordered $\mathrm{Aut}(G)$ classes. The reader checks these divisions before joining all 79 groups to all 88 source occurrences. The resulting complete machine-readable row list gives the stated total and sole discrepancy. The previously accepted independent literal 46-class certificate for $\mathrm{PSU}_4(2)$ corroborates this row without assuming exhaustion or outer-index division; it is acquired separately, not counted twice. □

The Section 6 aggregates. The accepted regular-column application gives 9210 regular classes, with rank distribution $8319, 810, 61, 16, 3, 1$ for ranks 3 through 8. Its only row correction is $\mathrm{PSL}_2(27)$, 27 to 26. Combining that result with the theorem gives the following family totals; overlapping group occurrences remain in their individual families.

Source table	Regular	Chiral	Regular rank ≥ 4	Chiral rank ≥ 4
2	1536	4780	271	360
3	804	1710	234	348
4	3903	1882	45	338
5	1846	3998	180	448
6	1202	4242	170	464
7	14	412	0	0

The nine duplicate occurrences contribute 95 regular and 50 chiral classes and are removed only in the global sum. The source's table-total arithmetic errors, already visible without a census, are Table 3 chiral $1702 \to 1710$, Table 4 chiral $2050 \to 1882$, and Table 5 regular $1840 \to 1846$. The two newly verified row corrections then give the global $9205 \to 9210$ regular and $17114 \to 16974$ chiral totals. The Section 6 comparison for the PSL_2 family has rank-at-least-four counts 45 regular and 338 chiral, rather than 51 and 506.

Mirror reversal is a fixed-point-free involution on the chiral ordered classes: a fixed class would provide the reflection automorphism excluded by chirality. Therefore identifying enantiomorphic forms gives $16974/2 = 8487$ classes, by rank $7528, 945, 14$. This does not identify duals. Under this convention regular classes still outnumber mirror pairs, whereas the two oriented forms counted separately outnumber regular classes. The qualitative sporadic, unitary and Suzuki comparisons in Section 6 remain true. Its older 5265 count is explicitly up to both isomorphism and duality, and the Conder genus counts are attributed historical context from other work; neither is inferred from this census. The source excludes $\mathrm{Aut}(J_2)$ from its regular aggregate, as do our 79 printed models. $J_2 : 2$ has order 1209600 and lies outside both numerical cutoffs discussed below.

Lemma 1 (Automorphisms of an almost simple preimage)**.** *Let S be nonabelian finite simple and $S \leq G \leq \mathrm{Aut}(S)$. Then restriction identifies*

$$\mathrm{Aut}(G) \cong N_{\mathrm{Aut}(S)}(G), \qquad N_{\mathrm{Aut}(S)}(G)/S = N_{\mathrm{Out}(S)}(G/S).$$

Proof. The centralizer of S in $\mathrm{Aut}(S)$ is trivial. Any nontrivial normal subgroup of G contains S: otherwise its commutator with S is trivial, which places it in that centralizer. Hence S is the unique minimal normal subgroup and is characteristic. An automorphism of G restricts to $\beta \in \mathrm{Aut}(S)$. Its value at $g \in G$, considered as an automorphism of S, must be $\beta g \beta^{-1}$, by conjugation on S. It therefore embeds in the displayed normalizer. Conversely a normalizer element induces an automorphism of G by conjugation. The same trivial centralizer makes this action faithful. The last equality follows since G is the full preimage of G/S. □

The group-order and automorphism-name columns. The accompanying 79-model registry rejoins every actual ordered generator array to its accepted named-model selector; natural alternating and symmetric models are identified by permutation parity and their checked full factorial order. These are identifications, not naming by order alone. Two printed group orders are wrong: M_{22} is 443520, not 443510, and $\mathrm{PSL}_2(25).2$ is 15600, not 7800.

For automorphism names we retain the full named simple-group automorphism classifications and the accepted outer-preimage identifications. The classified-input locators and the exact named target model for every row are recorded separately. Except for $L_3(4)$ and $U_3(5)$ the relevant outer groups are abelian, so every outer subgroup has full normalizer. In $D_{12} \cong \mathrm{Out}(L_3(4))$, the central order-two subgroup is normal, each reflection subgroup has a four-group normalizer, and that four-group is self-normalizing. The other relevant subgroups are normal. The already accepted ATLAS-word joins distinguish the central 2_1, field 2_2 and graph 2_3 types. Thus precisely the two reflection preimages and their four-group preimage

have automorphism group $L_3(4).2^2$; the others have the full $L_3(4).D_{12}$. In $S_3 \cong \mathrm{Out}(U_3(5))$ the reflection preimage $P\Sigma U_3(5)$ is self-normalizing; the other three preimages have full automorphism group $P\Gamma U_3(5)$. All these small normalizers are computed literally in the accompanying reader.

The source's $\mathrm{Aut}(\mathrm{PSU}_3(3)) = P\Gamma L_3(3)$ is a label error: the retained unitary automorphism theorem gives $P\Gamma U_3(3)$, of order 12096. In contrast, $P\Gamma L_3(3) = PGL_3(3)$ has order

$$\frac{(3^3-1)(3^3-3)(3^3-3^2)}{3-1} = \frac{11232}{2} = 5616,$$

since $\mathbb{F}_3$ has no nontrivial field automorphism. For J_2, the acquired full action is the accepted pair PrimitiveGroup(100, 1) inside PrimitiveGroup(100, 2). A saved six-letter word in the latter generators gives an actual outside involution; a clean bounded verification checks the exact arrays, normalization, index two and generation. Thus the printed colon $J_2 : 2$ has an actual complement, not merely an outer quotient of order two. The unsuccessful initial length-two proposal and the longer proposal's GAP parse warning are preserved; only the clean fixed-witness verification supplies this join.

Theorem 2 (Historical-domain boundary, with classified inputs)**.** *Using the retained CTblLib transcription of the 1985 ATLAS contents and the accepted complete simple/outer roster below one million, the Section 4 socle condition gives 80 almost simple isomorphism classes of order below one million. Exactly 79 have order below 900000, and their named models are precisely the 79 printed groups. The additional class is* $\mathrm{Sp}_4(4)$ *of order 979200.*

Proof. The accepted L054 domain theorem gives every almost simple group below one million as a preimage of an outer subgroup. Its unique-minimal-normal argument proves that isomorphisms between such preimages are exactly outer conjugacies. Its finite outer-subgroup lists and complete simple roster are acquired, with their explicit classification premises.

The pinned CTblLib `AtlasPages` record has 97 entries. Its own documentation identifies the original book's page-v contents and later additions. Exactly 93 entries have a nonempty ordinary-ATLAS page list; the other four are later additions $L_2(49), L_2(81), L_6(2), S_{10}(2)$. Retaining only the original ordinary-page entries intersects the accepted small-order roster in 32 socles and 80 almost simple classes. The literal table identifiers for the 79 acquired models are distinct and are exactly this list minus `sp44`. Their checked orders are all below 900000. The remaining group has order 979200 and ordinary ATLAS page 44. This proves both strict-cutoff assertions without guessing an old website edition or using polytope counts to identify groups. □

The precise source limit. The abstract on page 371 says that Γ has order less than 900000 and is listed in the ATLAS. Section 4, page 375, instead explicitly uses a socle S appearing there and $|\Gamma| < 10^6$. Our theorem implements that explicit socle condition. The 79 printed models agree with its 900000 restriction, but not with its one-million restriction. We have not inspected a scan of the 1985 contents: the pinned CTblLib transcription is an openly retained premise. We do not assert that every extension has its own separately printed 1985 table. The order boundary and this Γ-versus-S wording distinction are kept separate. No polytope census for $\mathrm{Sp}_4(4)$ is supplied by this component; no conclusion about an omitted zero count is licensed. Consequently this application closes the printed second-atlas entries and their aggregate corrections, but does not silently convert the larger Section 4 assertion into the narrower printed roster.

Attribution and limits. Root authored the prefix algorithm/theory, most original whole-group applications and the fixed queue, and suggested the normalizer lemma. The present author wrote the v2/v3 receipt repair, ran the four assigned fixed-plan queues, and assembled the present exact joins and small normalizer calculations. The corresponding independent reviews are acquired with these roles unchanged. The regular application and U4(2) positive correction have their separate independent reviews. Publication provided a bounded Section 6 source-only reading. No new polytope census, whole-manuscript acceptance, open-question resolution, Lean proof or compression measurement is asserted here.

Chiral polytopes by prefix coordinates

A complete finite enumeration contract

human_compression research project

11 October 2026 — author manuscript

Abstract

We recover the finite chiral-polytope enumeration of Buekenhout, Leemans and Tranchida through a shared theory of marked coordinates and exact orbit searches. The subgroup dictionary is literal: interval-rotation subgroups become difference subgroups without changing any intersection. We prove termination, completeness, the reflection test, and both oriented and unrooted equivalence conventions. Exact small-group enumerations validate the whole search, while uniform counts reconstruct the mathematical columns of the source timing table. The geometric reconstruction theorem of Schulte and Weiss and the almost-simple involution bound of Liebeck and Shalev remain attributed inputs. The source's runtime measurements are retained as empirical observations. The distinction between directly regular and nonorientable regular output is made explicit.

1 Source, scope and geometric input

Our principal source is the complete fourteen-page revised author manuscript *A new algorithm to classify chiral polytopes with a given automorphism group*, by Francis Buekenhout, Dimitri Leemans and Philippe Tranchida, deposited as `algo-rev.pdf` at the Université libre de Bruxelles. We denote it by BLT. Its journal citation is *Beiträge zur Algebra und Geometrie* **62** (2021), 21–36. The source ledger identifies the precise PDF bytes and separately records the earlier arXiv:1912.02287v1. The journal's sixteen-page typesetting has not been obtained, so a statement about the revised author text is not presented as a verified statement about every journal page.

The principal mathematical contract comprises BLT Theorem 4.3, Lemmas 4.4–4.5, and the complete search described in Section 4.2. We also recover the mathematical candidate-space comparison in Section 4.3, retaining its external involution bound. The only numerical table in BLT contains timings, group orders and involution proportions. It is not a table of numbers of chiral polytopes. Our exact small-group census is a validation of the recovered algorithm, not an invented reconstruction of an absent source census.

Throughout, isomorphisms preserve face ranks. Duality is a different equivalence and is not factored out. A connected Schläfli diagram means that every adjacent entry of its string symbol exceeds two. This is the nondegenerate scope stipulated in BLT Section 4. An *oriented* rotary polytope is supplied with a choice of one of the two parity classes of flags. Removing this choice gives the underlying unrooted polytope. In a chiral polytope these two choices are inequivalent under automorphisms of the polytope, but they have the same underlying partially ordered set.

We use the following classical reconstruction theorem, with its original attribution. It is Schulte–Weiss, *Chiral polytopes* (1991), Theorem 1, in the form restated in BLT Theorem 4.1 and Hartley–Hubard–Leemans (2012), printed pp.374–375. Let

$$G = \langle s_1, \dots, s_{n-1} \rangle, \qquad n \geq 3,$$

where $\mathrm{ord}(s_i) = p_i \geq 3$ and every consecutive product $s_i \cdots s_j$ with $i < j$ is a nonidentity involution. For $J \subseteq \{0, \ldots, n-1\}$ define

$$T_J = \langle s_{i+1} \cdots s_j : i < j,\ i, j \in J \rangle. \tag{1}$$

Suppose $T_J \cap T_K = T_{J \cap K}$ for all J, K. There is then a chiral or directly regular polytope of type $\{p_1, \ldots, p_{n-1}\}$ with rotation group G. It is directly regular precisely when there is an automorphism f satisfying

$$f(s_1) = s_1^{-1}, \quad f(s_2) = s_1^2 s_2, \quad f(s_i) = s_i \ (i \geq 3). \tag{2}$$

Its chain stabilizers in the rotation group are the groups in (1). Conversely, a chiral polytope with a chosen flag yields this marked data. These geometric existence and stabilizer assertions are retained, not reproved under the name of coordinate transport.

The exact-order formulation above uses a slightly stronger explicit input than the usual presentation by squared interval words. It holds for actual polytopes: the corresponding two distinct flag adjacencies cannot have product equal to the identity. It is also what the recovered enumeration tests. There is no silent assumption that arbitrary groups satisfying only the string relations are polytopal.

2 The complete subgroup dictionary

Put $a_0 = 1$ and $a_i = s_1 \cdots s_i$. For $J \subseteq \{0, \ldots, n-1\}$ put

$$D_J = \langle a_i^{-1} a_j : i, j \in J \rangle. \tag{3}$$

The shared marked-coordinate theorem MC.1 gives, simultaneously for every J,

$$s_i = a_{i-1}^{-1} a_i, \qquad s_{i+1} \cdots s_j = a_i^{-1} a_j, \qquad T_J = D_J. \tag{4}$$

Thus generation and the entire intersection diagram are preserved, rather than merely the top generated group. The empty and singleton subgroups in (3) are trivial; consequently the intersection property is automatic if either indexing set has at most one member. Adjoining the formal improper indices $-1, n$ with their associated trivial words changes no subgroup.

Theorem 1 (BLT prefix characterization, in ordered coordinates)**.** *Let G be finite. Its connected oriented chiral polytopes are represented by ordered normalized tuples $\boldsymbol{a} = (1, a_1, \ldots, a_{n-1})$ satisfying all of the following conditions:*

1. *$n \geq 3$ and $\langle a_1, \ldots, a_{n-1} \rangle = G$;*
2. *$\mathrm{ord}(a_{i-1}^{-1} a_i) > 2$ for $1 \leq i < n$;*
3. *$\mathrm{ord}(a_i^{-1} a_j) = 2$ whenever $0 \leq i < j < n$ and $j - i \geq 2$;*
4. *$D_J \cap D_K = D_{J \cap K}$ for all J, K;*
5. *no automorphism of G maps every a_i to a_i^{-1}.*

Two tuples represent the same oriented isomorphism class exactly when an automorphism of G sends the entries of one to the corresponding entries of the other.

Proof. By (4), conditions 1–4 are exactly the generating, order and intersection conditions in the retained reconstruction theorem. In particular a_i is an involution for $i \geq 2$, while a_1 has order greater than two. MC.5 identifies the automorphism in (2) with simultaneous inversion

of the a_i. Such an automorphism has square equal to the identity because its square fixes a generating set, and it is nonidentity because it inverts a_1. Thus condition 5 is precisely chirality.

A generating-marking isomorphism transports every subgroup (3), and hence the corresponding geometric construction and its chosen flag class. Conversely, an orientation-preserving polytope isomorphism may be followed by an automorphism in the transitive rotation group so that it sends a chosen base flag to the other chosen base flag. Its conjugation action then sends each distinguished rotation, and hence each prefix, to the corresponding one. This proves the exact oriented equivalence. An explicit flag-graph proof of the same statement is given in the next section. □

For completeness, the face-stabilizer formulas in BLT Proposition 3.4 follow directly from (1). Omitting index zero leaves $\langle s_2, \ldots, s_{n-1} \rangle$, and omitting index $n-1$ leaves $\langle s_1, \ldots, s_{n-2} \rangle$. Omitting an internal index i leaves

$$\langle \{ s_j : j \neq i, i+1 \} \cup \{ s_i s_{i+1} \} \rangle . \tag{5}$$

Every interval word avoiding endpoint i either avoids the crossing or crosses it using the paired factor $s_i s_{i+1}$, proving both inclusions in (5). Restriction to a section retains the corresponding contiguous rotation and subgroup data. The retained reconstruction theorem therefore gives the directly regular-or-chiral section statement of BLT Proposition 3.3. The fundamental connection between flag stabilizers and these subgroups remains the classical input.

3 Flag graphs, reflection and the missing mirror quotient

The flag graph of the retained construction has $2|G|$ vertices. A useful convention labels them $G \times \{+, -\}$ and sets

$$r_i(g, +) = (g a_i, -), \qquad r_i(g, -) = (g a_i^{-1}, +) \quad (0 \leq i < n). \tag{6}$$

This convention interchanges the two parity labels relative to the convention in which a_i is the right-action word $\rho_0 \rho_i$. To check the identification, represent the two cosets of the universal even subgroup by w and $w\rho_0$. Right multiplication by ρ_i sends these to $w a_i^{-1} \rho_0$ and $w a_i$, respectively. Exchanging the displayed parity labels gives (6). Thus (6) is the actual flag graph furnished by reconstruction, not an unrelated graph with the same order.

Each r_i is an involution without fixed points. Distant r_i, r_j commute because $(a_i^{-1} a_j)^2 = 1$. The graph is connected since the tuple generates G. Left multiplication by G acts on it, freely and transitively on each parity class.

Proposition 2 (Exact flag isomorphisms)**.** *Let $\boldsymbol{a}, \boldsymbol{b}$ be valid tuples of the same rank in finite groups of the same order. Every color-preserving isomorphism between their flag graphs, after a left translation in the target, is one of the following:*

$$\begin{aligned} &(g, \epsilon) \longmapsto (f(g), \epsilon), && f(a_i) = b_i; \\ &(g, \epsilon) \longmapsto (f(g), -\epsilon), && f(a_i) = b_i^{-1}, \end{aligned}$$

where f is a group isomorphism. Consequently, unrooted chiral polytopes are classified by the automorphism orbits of tuples followed by the extra identification

$$(a_1, \ldots, a_{n-1}) \sim (a_1^{-1}, \ldots, a_{n-1}^{-1}). \tag{7}$$

Every unrooted chiral class consists of exactly two oriented classes.

Proof. A connected bipartite graph has just two choices of its ordered bipartition. Hence an isomorphism either preserves or exchanges the displayed parity classes. Translate its image of $(1,+)$ to the appropriate vertex with first coordinate 1. Since color zero exchanges $(g,+)$ and $(g,-)$, the same map f occurs in both coordinates. For a parity-preserving map, preservation of color i gives $f(ga_i) = f(g)b_i$. For a parity-exchanging map it gives $f(ga_i) = f(g)b_i^{-1}$. MC.3, or its word argument, shows that f is the required group isomorphism. Conversely these equations verify every colored edge directly.

In particular, switching the parity labels is an isomorphism from the graph of $\boldsymbol{a}$ to the graph of $\boldsymbol{a}^{-1}$. The corresponding polytopes have the same unrooted isomorphism class. The two oriented classes coincide exactly when an automorphism of G inverts all entries. This is exactly the regularity condition excluded in Theorem 1. Therefore the mirror action is free on the chiral oriented classes, and each unrooted class has two of them. □

A color-preserving isomorphism of polytope flag graphs gives an isomorphism of the polytopes: an i-face is the connected component obtained by deleting color i, and incidence is witnessed by a common flag. Thus the graph calculation proves the claimed unrooted equivalence, not only a graph equivalence. It agrees with the explicit discussion of enantiomorphic base flags in Hartley–Hubard–Leemans, printed p.374.

The distinction has an immediate computational consequence. The final deduplication in the printed BLT code tests only a homomorphism sending a tuple to another tuple. It naturally identifies oriented classes. If "up to isomorphism" means unrooted poset isomorphism as defined in the source's Section 2, the additional step (7) is necessary. We provide both outputs. This is a convention correction with a reproducible finite witness, not a claim that the source's choice of prefixes loses chiral polytopes.

4 An exhaustive search in every rank

Here is a fully specified algorithm, independent of a computer algebra system's choice of generating sequence or conjugacy-class ordering. The input is a finite group with exact multiplication, inverse and equality operations.

1. List all nonidentity involutions $\mathcal{I}$. Choose one representative a_1 of every conjugacy class of elements of order greater than two.

2. At a prefix $(1, a_1, \ldots, a_k)$ compute its subgroup and all its difference subgroups (3), using the star formula MC.2.

3. If the prefix generates G, record it when $k \geq 2$ and do not extend it. Otherwise try every $x \in \mathcal{I}$ outside its generated subgroup. Require

$$\operatorname{ord}(a_k^{-1}x) > 2, \qquad \operatorname{ord}(a_i^{-1}x) = 2 \quad (0 \leq i < k), \tag{8}$$

   and test the full difference-subgroup intersection property on the enlarged prefix. Recurse exactly for the candidates that pass.

4. For each generating tuple, use MC.3 to test whether its simultaneous inverse tuple is isomorphic to it. Retain the tuples for which this test fails.

5. Remove duplicates using the canonical colored Cayley array of MC.3. This gives the oriented list. Pair the resulting list by (7) for the unrooted list. For either list report $\{\operatorname{ord}(a_{i-1}^{-1}a_i) : 1 \leq i < n\}$ in order.

Theorem 3 (Complete finite enumeration)**.** *The algorithm terminates for every finite input group. Its two outputs contain exactly one representative of every connected oriented, respectively unrooted, chiral polytope with full automorphism group isomorphic to the input group. No rank cutoff or list of possible Schläfli types is required.*

Proof. Every initial subtuple of a valid tuple satisfies the same order and difference-subgroup intersection tests in its generated subgroup. These properties are invariant under conjugation, so normalizing the first entry loses no tuple. The intersection identity applied to $\{0, \dots, k\}$ and $\{0, k+1\}$ shows that a valid next entry cannot belong to the old generated subgroup. It follows both that our exclusion of such entries is safe and that generation cannot occur before the end of a valid tuple.

MC.4 now proves termination and exhaustiveness of the prefix search. More explicitly, each extension at least doubles the order of the prefix subgroup and the first subgroup has order at least three. A rank-n recorded tuple therefore satisfies $3 \cdot 2^{n-2} \leq |G|$. The search tests all permitted new entries at every stage, so every normalized valid tuple occurs along one of its branches. Theorem 1 proves that the reflection filter retains exactly the chiral cases and that their full automorphism group is G. MC.3 gives precisely the oriented equivalence; Proposition 2 gives precisely the extra unrooted equivalence. This also proves the correctness of the reported types. □

The normalizer and centralizer pruning used by BLT follows directly from (8). A nonadjacent involutory entry x relative to a_1 satisfies $(a_1^{-1}x)^2 = 1$, equivalently $xa_1x = a_1^{-1}$, so it lies in the normalizer of $\langle a_1 \rangle$. With an earlier involution a_i, the condition $(a_ix)^2 = 1$ is equivalent to commutation; the order is exactly two because the entries are distinct. For the immediately previous involution, order greater than two is equivalent to noncommutation. Restrictions inherited from all earlier nonadjacent positions must be retained when the next entry is tried; the restriction from the preceding adjacent position must not be incorrectly retained after it becomes nonadjacent. Direct order tests avoid this bookkeeping issue. Stabilizer-orbit pruning is optional and is justified by U.ExtensionOrbits as incorporated in the companion paper, Section 2.

Our implementation uses all candidates after the first-entry conjugacy reduction. It does not rely on the order in which software lists conjugacy classes. It tests the first adjacent order explicitly, and it uses the supplied ordered tuple rather than asking whether an implicit computer algebra generating sequence happens to have the intended order. These choices produce a transparent reference implementation; they are not a claim to improve the original optimized runtime.

5 Candidate counts and the source timing table

Write $N = |G|$, $v = |\mathcal{I}|$, and $u = N - 1 - v$, the number of elements of order greater than two. Before subgroup, order or conjugacy pruning, the original rank-n rotation search has at most u^{n-1} tuples, while the prefix search has at most

$$u\, v^{n-2}. \tag{9}$$

Indeed only its first nonidentity coordinate is selected from the u noninvolutions, and all remaining coordinates are involutions. If $u > 0$, the ratio of these Cartesian candidate counts is $(v/u)^{n-2}$. Distinctness, conjugacy representatives and prefix rejection can only reduce the counts. Equation (9) compares candidate sets; it is not a worst-case ratio between complete running times, since individual subgroup and isomorphism tests also have costs.

BLT Proposition 4.6 retains the Liebeck–Shalev involution estimate: for almost simple groups outside the family with socle $\mathrm{PSL}_2(q)$, there is an absolute constant c with $v < cN^{5/8}$.

We retain precisely this attributed input as quoted in BLT; its classification-theoretic proof is not included in either of our two papers. It implies $v/N = O(N^{-3/8})$, and hence $v/u = O(N^{-3/8})$ along such a family. Thus (9) explains the decreasing candidate proportion at each fixed rank. No uniform small numerical value is implied for every small group by an unspecified constant.

The timings in the source were measured on the stated machine with Magma. They are empirical records and cannot follow as corollaries of an abstract proof. We do not recreate their hardware or claim to reproduce those timings. Their group-order and involution-proportion columns, however, have exact mathematical content. The following uniform formulas reconstruct them.

Proposition 4 (Exact involution counts for all table families)**.** *Let q be a prime power. Then*

$$|\operatorname{PSL}_3(q)| = \frac{q^3(q^3-1)(q^2-1)}{(3,q-1)}, \qquad i_2(\operatorname{PSL}_3(q)) = \begin{cases} q^2(q^2+q+1) & q \text{ odd}, \\ (q^3-1)(q+1) & q \text{ even}; \end{cases}$$

$$|\operatorname{PSU}_3(q)| = \frac{q^3(q^3+1)(q^2-1)}{(3,q+1)}, \qquad i_2(\operatorname{PSU}_3(q)) = \begin{cases} q^2(q^2-q+1) & q \text{ odd}, \\ (q^3+1)(q-1) & q \text{ even}. \end{cases}$$

Also

$$i_2(S_n) = \sum_{k=1}^{\lfloor n/2 \rfloor} \frac{n!}{2^k k!(n-2k)!}, \qquad i_2(A_n) = \sum_{\substack{1 \le k \le n/2 \\ k \text{ even}}} \frac{n!}{2^k k!(n-2k)!}. \tag{10}$$

Proof. Counting ordered bases gives $|\operatorname{GL}_3(q)| = (q^3-1)(q^3-q)(q^3-q^2)$. Divide by the $q-1$ determinant values and then by the center of order $(3,q-1)$ to obtain the first group order. In odd characteristic an involution in $\operatorname{SL}_3(q)$ has a one-dimensional $+1$ eigenspace and a complementary two-dimensional -1 eigenspace. There are q^2+q+1 choices of the line and q^2 complements of it. Conversely every such ordered decomposition determines one involution. In characteristic two, an involution is $1+u\varphi$, where $u \neq 0$, $\varphi \neq 0$, and $\varphi(u) = 0$. Its nilpotent part has rank one because its square is zero in dimension three. For each image line, fix one nonzero u; there are q^2-1 suitable nonzero functionals. Thus the count is $(q^2+q+1)(q^2-1) = (q^3-1)(q+1)$.

For the unitary calculation use a nondegenerate Hermitian form on $\mathbb{F}_{q^2}^d$, and let z_d be the number of norm-zero vectors. Each nonzero value of the field norm has $q+1$ preimages, while zero has one. Hence $z_{d+1} = (q+1)q^{2d} - qz_d$, starting with $z_1 = 1$. This gives $z_2 = q^3+q^2-q$ and $z_3 = q^5-q^3+q^2$. There are consequently q^3+1 isotropic lines and $q^2(q^2-q+1)$ nonsingular lines in dimension three. The counts of unit vectors in dimensions 1, 2 and 3 are, respectively, $q+1$, $q(q^2-1)$ and $q^2(q^3+1)$. Counting orthonormal bases gives $|\operatorname{GU}_3(q)| = q^3(q^3+1)(q^2-1)(q+1)$; the determinant map has $q+1$ values, giving the stated order of $\operatorname{SU}_3(q)$ before quotienting its center.

In odd characteristic, an involution in $\operatorname{SU}_3(q)$ is $+1$ on a nonsingular line and -1 on its orthogonal plane. Thus the number is exactly the number of nonsingular lines. In characteristic two its nilpotent part again has rank one. If its image is $\langle u \rangle$, preservation of the form implies that its kernel is $u^\perp$. The inclusion of image in kernel makes u isotropic. It therefore has the form $N(x) = c\,h(x,u)u$, with $c + c^q = 0$. Conversely this formula gives a determinant-one unitary involution for every nonzero such c. For a fixed nonzero representative u of an isotropic line there are $q-1$ choices, giving $(q^3+1)(q-1)$.

In both special groups the center has odd order dividing three. Every projective involution has a unique involutory lift: if x^2 is central, multiplication by the unique central square root of x^{-2} produces the lift. The projective counts therefore equal the counts just obtained. Finally a permutation involution consists of k disjoint transpositions; selecting and pairing their $2k$ moved points gives the summand in (10). It is even precisely when k is even. □

The accompanying exact table records all 32 rows of the revised author source. Twenty of the counts are checked independently by GAP conjugacy classes: both degree-three classical families for $q = 2, 3, 4, 5$, and both symmetric and alternating families for $5 \le n \le 10$. The all-parameter conclusion is the written proof, not an extrapolation from these twenty checks. For example, the revised source's unitary ratios 0.053 for $\mathrm{PSU}_3(2)$ and 0.0097 for $\mathrm{PSU}_3(4)$ differ from the exact values $9/72 = 1/8$ and $195/62400 = 1/320$. Its $\mathrm{PSU}_3(5)$ ratio is $525/126000 = 1/240$. The revised source already corrects the earlier arXiv entry for $\mathrm{PSL}_3(17)$; that earlier error is not attributed to the revised text.

6 Exact models and the regular-output boundary

The reference enumeration includes the trivial group and elementary abelian groups of orders two and four as empty-output boundary cases. It constructs finite multiplication tables from explicit permutations and searches every rank allowed by the subgroup-growth proof. Its counts are as follows. The regular column counts connected directly regular polytopes with the indicated group as rotation group, with duals kept distinct.

Input group	Oriented chiral	Unrooted chiral	Directly regular
C_5	0	0	0
D_{10}	0	0	0
A_4	0	0	1
S_4	0	0	2
A_5	0	0	4
$\mathrm{AGL}_1(5)$	2	1	0
$\mathrm{AGL}_1(7)$	4	2	0
$\mathrm{PSL}_2(7)$	0	0	5

For A_5 the four types are $\{3,5\}, \{5,3\}, \{5,5\}$ and the rank-four type $\{3,3,3\}$. For $\mathrm{AGL}_1(5)$ the chiral type is $\{4,4\}$; for $\mathrm{AGL}_1(7)$ they are $\{3,6\}$ and $\{6,3\}$. An independent rank-three search tries every ordered pair of original rotations, verifies generation and all three cyclic subgroup intersections, and agrees with every rank-three prefix output. The saved witnesses also undergo a complete color-preserving graph-automorphism search and the component-intersection tests at both flag orbits. The mirror isomorphism in (7) is checked on every colored edge.

The group of order 20 can be written $\mathbb{F}_5 \rtimes C_4$, with multiplication $(u, i)(v, j) = (u+2^i v, i+j)$. Take $a_1 = (0, 1)$ and $a_2 = (1, 2)$. Their successive rotation orders are 4 and 4, and the three relevant cyclic subgroups have pairwise trivial intersections. They generate the whole group. Its characteristic derived subgroup is the translations. An automorphism conjugates the action of the quotient on this subgroup through an automorphism of C_5; that automorphism group is abelian, so the quotient action $u \mapsto 2u$ cannot be changed to $u \mapsto 2^{-1}u$. Thus no automorphism inverts a_1. This gives the explicit chiral mirror-pair example without relying solely on a census count.

There are two further boundary controls. First, the tetrahedral A_4 marking is inverted by an outer automorphism, but not by any inner automorphism. Replacing $\mathrm{Aut}(G)$ by G in the reflection test would falsely declare it chiral. This is also why the automorphism-language repair already present in the revised BLT Lemma 4.4 must be preserved.

Second, the final paragraph of BLT says that removing the chirality test also computes all regular polytopes when G is simple. The retained reconstruction theorem instead returns directly regular polytopes with rotation group G; these have $2|G|$ flags and full group of order $2|G|$. A regular polytope whose full group is the nonabelian simple group G has $|G|$ flags. Its rotation subgroup is G because G has no subgroup of index two, so it is nonorientable and is not the same reconstructed object.

An explicit certificate uses A_5 on $\{0,1,2,3,4\}$ with

$$r_0 = (1\,3)(2\,4), \qquad r_1 = (0\,3)(2\,4), \qquad r_2 = (1\,4)(2\,3). \tag{11}$$

These generate A_5, the end generators commute, their consecutive product orders are 3 and 5, and all 64 standard subgroup-intersection tests pass. Thus they define a regular 60-flag polytope. Their prefix rotation tuple $(1, r_0r_1, r_0r_2)$ generates A_5 and passes the difference-subgroup tests. Applying the rotary reconstruction gives a directly regular 120-flag polytope. The two flag counts exclude isomorphism. Consequently the last paragraph needs an additional nonorientable quotient or an independent ordinary C-group search; it does not follow by simply deleting the filter. The chiral enumeration theorem is unaffected.

7 Source clauses, retained costs and verification limits

The source ledger assigns each principal and supporting clause to a specific proof span in this TeX source or in *Marked coordinates and exact orbit enumeration*. The latter paper is a reusable acquisition; this application pays separately for the geometric interface, the actual admissibility and equivalence arguments, and the exact table calculations. The source's use of the Schulte–Weiss theorem and the Liebeck–Shalev bound is retained with attribution. The prefix idea, the predominance of involutory candidates and the reflection test are the source authors' contributions and inputs, not credited to the model as inventions.

The revised text still has transcription inconsistencies. Its equation (6) includes $i = j$ in the squared interval relation, which would force each adjacent rotation to have order at most two; it must read $i < j$ in this use. Equation (10) gives distance two in both cases; the non-involutory case must have distance one. Theorem 1 states the consistent conditions explicitly. The index formulas in its discussion of A_I and the piecewise expansion inside Theorem 4.3 are replaced by the exact dictionary (4). We distinguish these corrections from the already corrected automorphism language and $\mathrm{PSL}_3(17)$ entry in the revised edition. The omitted explicit first adjacent-order check in the printed recursion is not used to assert that a chiral output is necessarily wrong; our implementation simply tests every condition in its declared contract.

The seven Lean declarations in `PrefixCoordinates.lean` verify the all-length coordinate cancellation, its two inverses, the equality of actual subgroup closures, intersection transport, the reflected second generator, and the involutivity of a simultaneous-inversion automorphism on a generating set. They use only standard reported axioms. They do not formalize the Schulte–Weiss polytope construction, the whole search, the Lie-theoretic input bound, or the classification table calculations. The finite scripts and negative controls supplement the written proof; they do not replace its arbitrary-group and arbitrary-rank quantifiers.

No new conjecture is stated in this source's algorithm sections, and no open-question solution is claimed by this recovery. The finite corrections and equivalence clarification are kept in the source contract rather than assigned an artificial infinite compression ratio. A later matched measurement must charge the full relevant umbrella and the full retained-input boundary, together with the application. This paper makes no numerical compression or performance claim.

References

[1] F. Buekenhout, D. Leemans and P. Tranchida, *A new algorithm to classify chiral polytopes with a given automorphism group*, Beitr. Algebra Geom. **62** (2021), 21–36. Exact principal edition: ULB revised author manuscript, `https://dipot.ulb.ac.be/dspace/bitstream/2013/315729/3/algo-rev.pdf`; comparison edition: arXiv:1912.02287v1. DOI: 10.1007/s13366-020-00536-1.

[2] M. I. Hartley, I. Hubard and D. Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Math. Contemp. **5** (2012), 371–382; especially pp.374–375 for the retained reconstruction and enantiomorphic base flags.

[3] E. Schulte and A. Ivić Weiss, *Chiral polytopes*, DIMACS Ser. Discrete Math. Theoret. Comput. Sci. **4** (1991), 493–516, Theorem 1. The exact retained statement is read in BLT and HHL; the full original chapter was not obtained in this audit.

[4] M. W. Liebeck and A. Shalev, *Classical groups, probabilistic methods, and the* $(2,3)$*-generation problem*, Ann. of Math. **144** (1996), 77–125, Corollary 4.5 as retained in BLT Proposition 4.6. Its proof is not claimed recovered here.

[5] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Highly symmetric hypertopes*, Aequationes Math. **90** (2016), 1045–1067. This is the source of the C^+-group viewpoint used by BLT; our string application uses the classical polytope reconstruction directly.

[6] human_compression research project, *Marked coordinates and exact orbit enumeration*, companion TeX manuscript, phase 282. Its orbit interface reuses U.ExtensionOrbits from the shared proof bank and accepted enumeration applications.

Marked coordinates and exact orbit enumeration

human_compression research project

11 October 2026 — author manuscript

1 Coordinates on a marked group

A marking is an ordered tuple, not an unordered generating set. The distinction matters whenever the positions specify face ranks. Let G be a group and let $\boldsymbol{s} = (s_1, \dots, s_m)$ be a tuple in G. Define

$$a_0 = 1, \qquad a_i = s_1 \cdots s_i \quad (1 \le i \le m). \tag{1}$$

The reverse substitution is $s_i = a_{i-1}^{-1} a_i$. For a set $J \subseteq \{0, \dots, m\}$ put

$$D_J(\boldsymbol{a}) = \langle a_i^{-1} a_j : i, j \in J \rangle, \quad T_J(\boldsymbol{s}) = \langle s_{i+1} \cdots s_j : i < j,\ i, j \in J \rangle. \tag{2}$$

An empty generating set generates the trivial subgroup.

Theorem MC.1 (coordinate transport). The substitutions in (1) are mutually inverse, preserve the generated group, commute with every group homomorphism, and satisfy

$$a_i^{-1} a_j = s_{i+1} \cdots s_j \quad (i < j), \qquad D_J(\boldsymbol{a}) = T_J(\boldsymbol{s}). \tag{3}$$

Consequently every assertion involving these particular subgroup diagrams, their intersections, their generated joins and their element orders can be transferred by substitution, with no new group-theoretic hypothesis.

Proof. Adjacent factors cancel in $(a_i^{-1} a_{i+1}) \cdots (a_{j-1}^{-1} a_j)$, leaving $a_i^{-1} a_j$. Conversely, cancellation in $(s_1 \cdots s_i)^{-1} (s_1 \cdots s_j)$ gives the first equality in (3). Each original generator is a word in the new ones and conversely, so generation is preserved. A homomorphism commutes with products and inverses. In (2), a difference with reversed endpoints is the inverse of the difference with increasing endpoints, and equal endpoints contribute the identity. Thus the two displayed generating sets generate exactly the same subgroup. All the asserted transports now follow by replacing equal subgroups or equal elements. □

This elementary substitution is the marked-coordinate principle already recorded as M2 in the project proof bank. Its application to chiral polytopes is the substitution of Buekenhout–Leemans–Tranchida, Theorem 4.3; neither that substitution nor its use of C^+-groups is claimed as new here. The formal certificate accompanying this paper verifies (3) for arbitrary interval lengths and the equality of the actual subgroup closures, including reversed endpoint pairs.

Lemma MC.2 (a star generates the difference subgroup). For nonempty J and $b \in J$,

$$D_J(\boldsymbol{a}) = \langle a_b^{-1} a_j : j \in J \setminus \{b\} \rangle. \tag{4}$$

Proof. The right side is contained in the left. Every generator of the left has the expression $a_i^{-1} a_j = (a_b^{-1} a_i)^{-1} (a_b^{-1} a_j)$, proving the reverse inclusion. □

Formula (4) reduces a subgroup computation with quadratically many named differences to one with at most $|J| - 1$ generators. It changes neither the subgroup nor the intersection tests.

2 Exact equivalence of generating markings

We recall the shared orbit interface used in the enumeration topic. If a group B acts on sets Z, E, and $p : Z \to E$ is equivariant, then

$$B\backslash Z \;\simeq\; \coprod_{e\in B\backslash E} B_e\backslash p^{-1}(e), \tag{5}$$

where one base representative is chosen in each orbit. This is U.ExtensionOrbits. The proof and its previous geometric applications are retained from the shared umbrella: first move the base into the selected orbit representative; any remaining equivalence fixes that base. Formula (5) applies to a tuple and its initial subtuple. In particular, after fixing one entry one must use its stabilizer, rather than continuing to quotient the remaining entries independently by the whole group.

For a finite group G, let $\mathcal{M}_m(G)$ be the set of ordered generating m-tuples. The automorphism group acts freely on this set: an automorphism fixing all the generators fixes every word. Thus every orbit has size $|\operatorname{Aut}(G)|$. One need not construct the whole automorphism group to test this equivalence.

Theorem MC.3 (Cayley transport). Let $\boldsymbol{a}$ and $\boldsymbol{b}$ be generating m-tuples of finite groups G, H of equal order. The assignment $a_i \mapsto b_i$ extends to a group isomorphism if and only if the following finite propagation succeeds. Start with $f(1) = 1$, and whenever $f(x)$ is assigned, require

$$f(xa_i) = f(x)b_i \quad (1 \le i \le m). \tag{6}$$

Reject if two requirements disagree. Accept if the resulting assignment is a bijection. The isomorphism, when it exists, is unique.

Proof. Every group isomorphism with the required generator images satisfies (6), so disagreement or failure of bijectivity is a valid rejection. A finite group generated as a group by the a_i is also generated as a monoid: a_i^{-1} is a positive power of a_i. Hence propagation visits all of G. If it succeeds, induction on a positive word w gives $f(xw(\boldsymbol{a})) = f(x)w(\boldsymbol{b})$. Taking $x = 1$ shows that the right side depends only on the element represented by w. Taking a word for an arbitrary y then gives $f(xy) = f(x)f(y)$. Acceptance therefore gives a bijective homomorphism. Uniqueness follows from generation. □

A canonical implementation starts at the identity, visits new elements in breadth-first order using the ordered generators, and records the target numbers of all colored Cayley edges. Two such finite arrays agree exactly when the markings are isomorphic. Indeed equality of arrays supplies the bijection in MC.3; conversely an isomorphism carries the deterministic traversal to the same traversal. This test does not quotient the tuple by a permutation of its positions.

The same method applies to connected graphs with one edge of each color at every vertex: a color-preserving graph map is forced by the image of one vertex. Trying every image, rejecting inconsistencies and testing bijectivity enumerates the full color-preserving automorphism group. This is a direct finite certificate, not a heuristic based on a chosen subgroup of graph automorphisms.

3 Hereditary searches and termination

Suppose an admissibility predicate is defined on normalized tuples $(1, a_1, \dots, a_m)$ and has the following properties:

1. every initial subtuple is admissible in the subgroup it generates;
2. admissibility is preserved by automorphisms of the ambient group;
3. every appended entry lies in a prescribed finite candidate set;

4. every admissible extension enlarges the subgroup generated by the prefix.

Theorem MC.4 (exhaustive prefix search). Start at representatives of the allowed first-entry conjugacy classes. Recursively append every candidate that passes the admissibility tests. Record a tuple when it generates the input group, and stop that branch. Deduplication by MC.3 gives exactly the automorphism orbits of all admissible generating tuples.

Proof. Every retained tuple is admissible by the tests, and is recorded only when it generates the required group. Conversely, conjugate any admissible generating tuple to put its first entry into the chosen conjugacy representative. Every prefix passes the tests by hypothesis 1; induction on its length shows that the recursion reaches the entire tuple. Hypothesis 4 shows that no proper prefix already generates the ambient group, so stopping a branch early loses no output. The search terminates because each subgroup index along a strict finite subgroup chain is at least two. Finally MC.3 removes exactly, and all, the automorphism equivalences among the recorded tuples. □

The conjugacy reduction need not remove every automorphism equivalence at the first entry. Inner automorphisms suffice for completeness, and final automorphism deduplication removes any remaining duplicates. Alternatively, (5) permits successive reduction by the actual stabilizer of each prefix. Such reduction is optional; its implementation must preserve the prefix rather than treating entries independently.

A useful sufficient condition for strict growth is the difference-subgroup intersection property

$$D_J \cap D_K = D_{J\cap K}. \tag{7}$$

For a prefix $(1, a_1, \ldots, a_{k-1})$ and a nonidentity new entry a_k, take $J = \{0, \ldots, k-1\}$ and $K = \{0, k\}$. Then $\langle a_1, \ldots, a_{k-1}\rangle \cap \langle a_k\rangle = 1$. In particular a_k is not in the old subgroup. If $|a_1| \geq 3$ and the final tuple has n coordinates including a_0, then

$$|G| \geq 3\,2^{n-2}. \tag{8}$$

The implementation may use this bound or simply stop whenever generation occurs. Its proof of termination is independent of a numerical cutoff selected by the programmer.

4 Reflection tests

Let $(1, a_1, \ldots, a_{n-1})$ generate G, assume $a_i^2 = 1$ for $i \geq 2$, and write $s_i = a_{i-1}^{-1} a_i$.

Proposition MC.5 (reflection transport). An automorphism f satisfies

$$f(a_i) = a_i^{-1} \quad (1 \leq i < n) \tag{9}$$

if and only if

$$f(s_1) = s_1^{-1}, \qquad f(s_2) = s_1^2 s_2, \qquad f(s_i) = s_i \quad (i \geq 3). \tag{10}$$

Any such automorphism has square equal to the identity. If $|a_1| > 2$, it is not the identity.

Proof. Condition (9) fixes a_i for $i \geq 2$. Therefore $f(s_2) = f(a_1^{-1}a_2) = a_1 a_2 = a_1^2 s_2$, and both factors in $s_i = a_{i-1}^{-1} a_i$ are fixed for $i \geq 3$. This gives (10). Conversely (10) sends the first prefix to its inverse and fixes the second prefix $s_1 s_2$. All subsequent prefixes are then fixed. These are exactly (9), since they are involutions. Finally f^2 fixes every generating entry, and hence is the identity. Its action on a_1 is nontrivial when $|a_1| > 2$. □

This is the content of BLT Lemma 4.4 in the correct automorphism language, expressed through MC.1. It concerns all group automorphisms, not only conjugations by elements of G. MC.3 decides (9) without requiring the automorphism group to be enumerated in advance.

5 Attribution, formal coverage and reuse

The umbrella collects elementary coordinate and enumeration interfaces. MC.1 is the existing marked-coordinate principle; the specific prefix substitution and reflection transport used here belong to BLT. The orbit-over-a-base result (5) is U.ExtensionOrbits, already used in the accepted enumeration module. MC.2, MC.3 and MC.4 supply explicit complete proofs of the algebraic and algorithmic details needed by applications; no priority is asserted for these elementary statements.

`PrefixCoordinates.lean` proves seven declarations: arbitrary-length telescoping, recovery in both coordinate directions, equality of subgroup closures, transport of the full intersection assertion, the reflected second generator, and the square of an inversion automorphism on an arbitrary generating set. The subgroup theorem is not merely equality of two generated groups at the top: it identifies every subgroup indexed by a set of endpoints. The formal file does not prove the geometric reconstruction theorem, finite search completeness, the involution-count theorem for almost simple groups, or the correctness of the Python interpreter and GAP.

The associated application must pay for its geometric recognition theorem and for its specific admissibility conditions. This umbrella is charged in full to any topic using it under the project's requested topic-local measurement convention. Shared acquisition inside one topic is recorded once; no numerical compression ratio is claimed here.